Robot Perception and Learning

Intelligent Exploration

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Exploration is the Core Problem in Reinforcement Learning

- The goal of reinforcement learning is to maximize the return
- Exploration: the agent experiments with novel strategies that may improve returns in the long run
- Exploitation: the agent maximizes rewards through behavior that is known to be successful
- Exploitation makes sense only when the agent had stumbled into high-reward states
- What if the reward is extremely sparse (e.g. Success vs. Failure)?

Easy env



Hard env



A Naïve Solution: Shaping Rewards

- Rewards in an easy env: success / Failure
- Rewards in a hard env: success / Failure, proximity to the object, overlapping of the object and the target
- Rewards in a very hard env: success / Failure, proximity to the object, overlapping of the object and the target, distance to the barrier, and MORE

Easy env



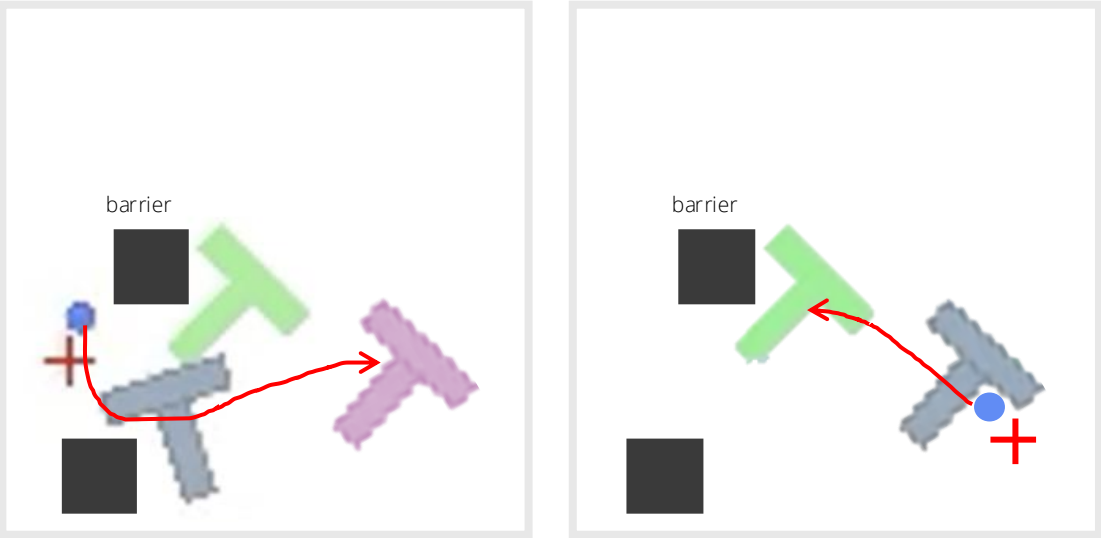
Hard env



How about this?



Reward Shaping Helps to Discover Intended Behaviors



- Intended behaviors:
  - Avoid the barrier
  - Move the object to spacious regions
  - Move the object to the target location

Shaping Rewards Require Prior Knowledge on the Task

Montezuma's revenge



- Getting key = reward
- Opening door = reward
- Getting killed by skull = nothing (is it good? bad?)
- Finishing the game only weakly correlates with rewarding events
- We know what to do because we **understand** what these sprites mean!

Slide adapted from S. Levine

Shaping Rewards Require Prior Knowledge on the Task
What if the Task is Extremely Complex…



Mao

- "the only rule you may be told is this one"
- Incur a penalty when you break a rule
- Can only discover rules through trial and error
- Rules don't always make sense to you

- Temporally extended tasks like Montezuma's revenge become increasingly difficult based on
  - How extended the task is
  - How little you know about the rules
- Imagine if your goal in life was to win 50 games of Mao…
- (and you didn't know this in advance)

Slide adapted from S. Levine

Can We Learn to Discover Intended Behaviors without / with Least Reward Shaping?



Image credit S. Levine

- Babies are excellent explorers. They try out different body poses. They play with different objects.
  - Learn skills unsupervisedly
  - Learn the dynamics of the environment
- What are the factors that drive babies to explore?
- In other words, how can we do exploration more efficiently and effectively?

Intrinsic vs. Extrinsic Motivations

- Motivation: “Forces” that energize an organism to act and that direct its activity

- Extrinsic Motivation: being moved to do something because of some external reward ($$, a prize, etc.).
  - Problem: such rewards are sparse..

- Intrinsic Motivation: being moved to do something because it is inherently enjoyable (curiosity, exploration, novelty, surprise, incongruity, complexity…)
  - Gain: Task independent! Free of human supervision, no need to code up reward functions to incentivize the agent. A general loss functions that drives learning.

Slide credit K. Fragkiadaki

Intrinsic Motivation vs. Intrinsic Necessity

- Curiosity vs. Survival

"As knowledge accumulated about the conditions that govern exploratory behavior and about how quickly it appears after birth, it seemed less and *less likely that this behavior could be a derivative of hunger, thirst, sexual appetite, pain, fear of pain, and the like*, or that stimuli sought through exploration are welcomed because they have previously accompanied satisfaction of these drives."

D. E. Berlyne, *Curiosity and Exploration*, Science, 1966

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How to Model Exploration?

- We can re-formulate the objective of reinforcement learning:

$$R^t(s, a, s') = r(s, a, s') + \mathcal{B}^t(s, a, s')$$

extrinsic reward (related to the task)

intrinsic reward (unrelated to the task)

- Types of intrinsic rewards:
  1. Data-based exploration (Optimistic exploration)
  2. Knowledge-based exploration
  3. Skill-based exploration
  4. Reachability exploration
  5. Posterior sampling for exploration

Data-based Exploration

- Idea: Maximize the coverage and minimize the uncertainty of data distribution
- For example, deciding the exploration bonus based by counting the times of visiting the action / state
  - ➢ Upper Confidence Bounds (UCB)

$$A_t \doteq \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\log t}{N_t(a)}} \right]$$

- t: how many times I have played any action,
- $N_t(a)$: how many times I have played action a in t interactions

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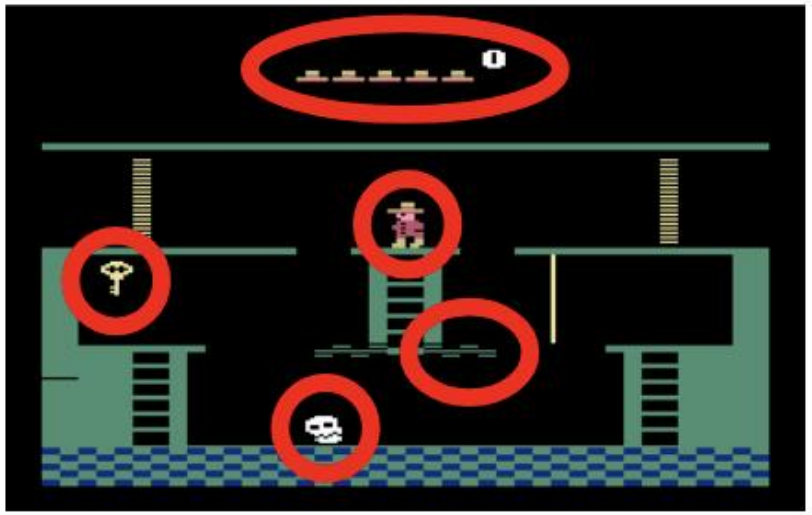
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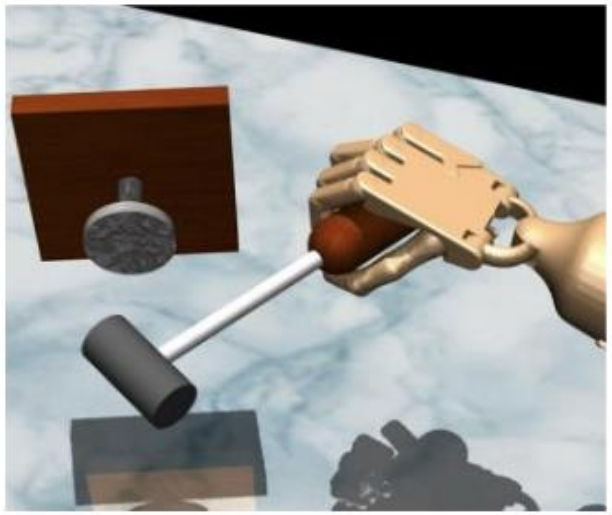
- We can count the times of visiting the state instead (select the action that goes to the least-frequently visited state)
- A strong assumption of discrete action / state space

- t: how many times I have played any action,
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The Trouble with Counts

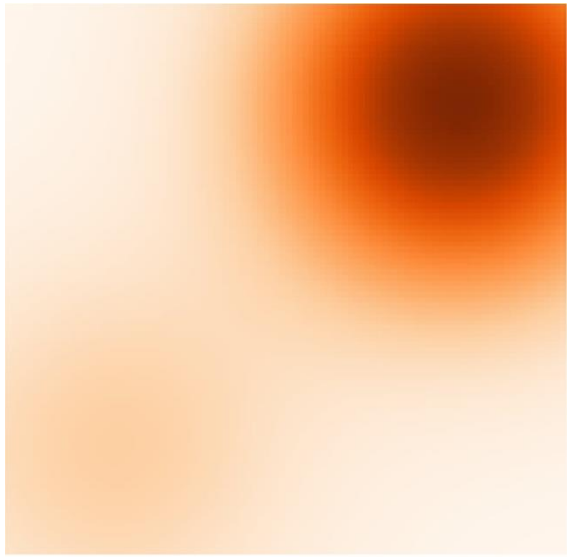
- How to count continuous states/actions?
  - ➢ We never see the same thing twice
  - ➢ Some states are more similar than others





Slide credit S. Levine

Idea 1: Fitting Generative Models



- Fit a generative model that captures the distribution of seen data $p_\theta(x)$ (e.g. diffusion models, energy-based models …)
  - ➢ $p_\theta(x)$ is high for seen data or unseen but similar data

- Can we use $p_\theta(x)$ to get a "pseudo-count"?

The true probability:

$$P(x) = \frac{N(x)}{n}$$

$N(x)$: count of x

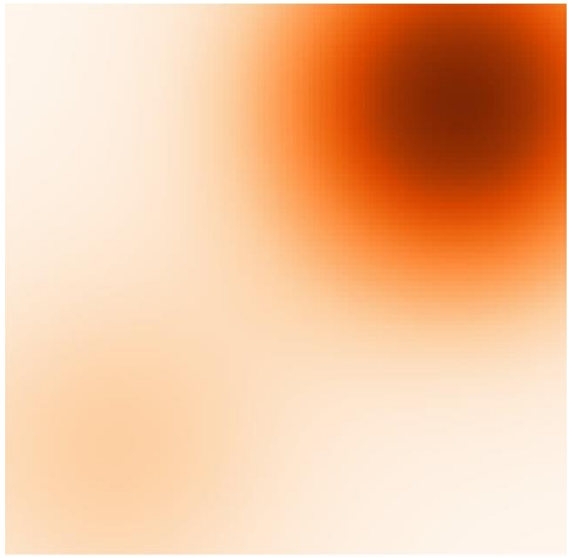
n: total count

After seeing x:

$$P'(x) = \frac{N(x)+1}{n+1}$$

Slide adapted from S. Levine

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pseudo count of x

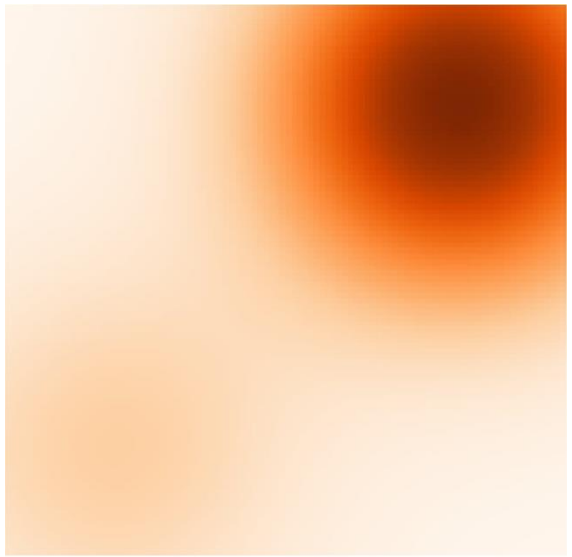
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$$\hat{N}(x) = \hat{n}\, p_\theta(x)$$

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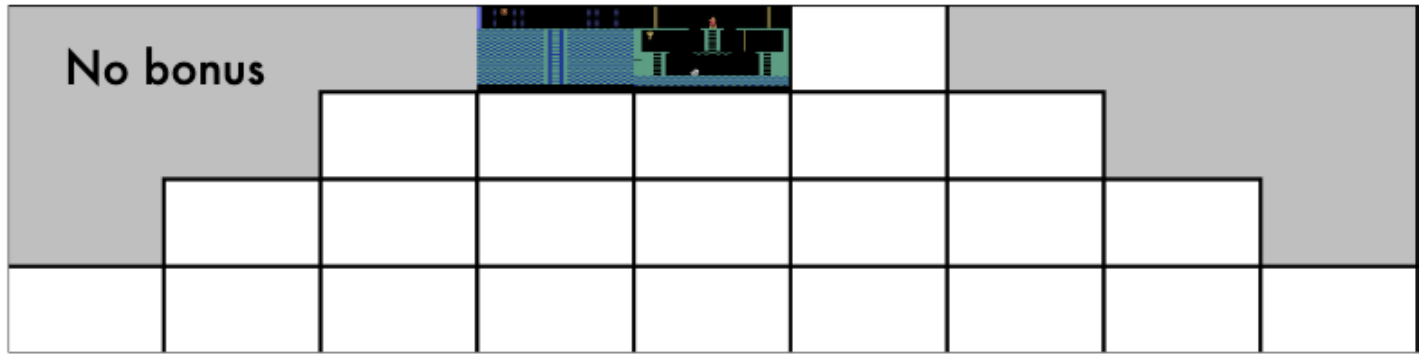
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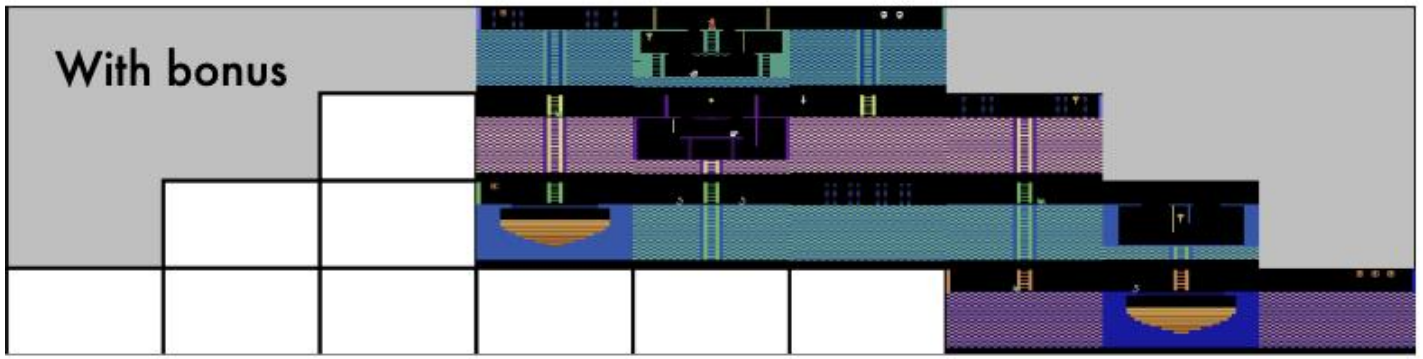
$$\hat{n} = \frac{1 - p_{\theta'}(x)}{p_{\theta'}(x) - p_\theta(x)}\, p_\theta(x)$$

Unifying Count-Based Exploration and Intrinsic Motivation. Bellemare et al.

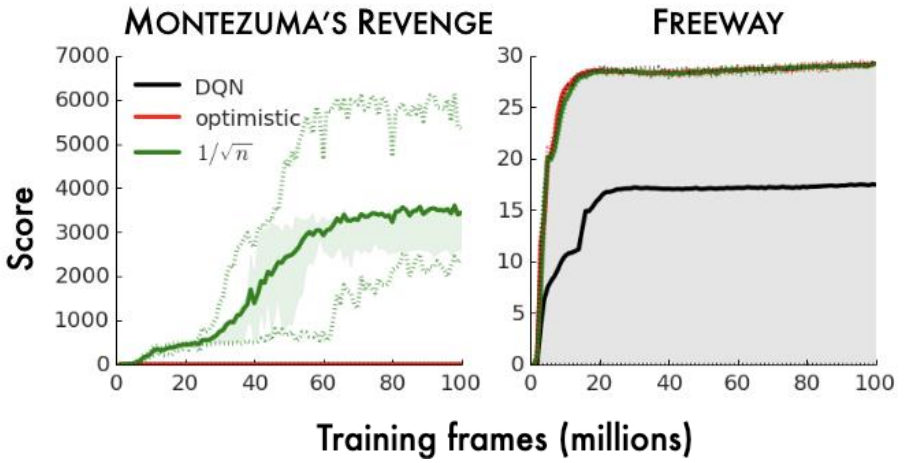
Slide adapted from S. Levine

Does It Work?


No bonus

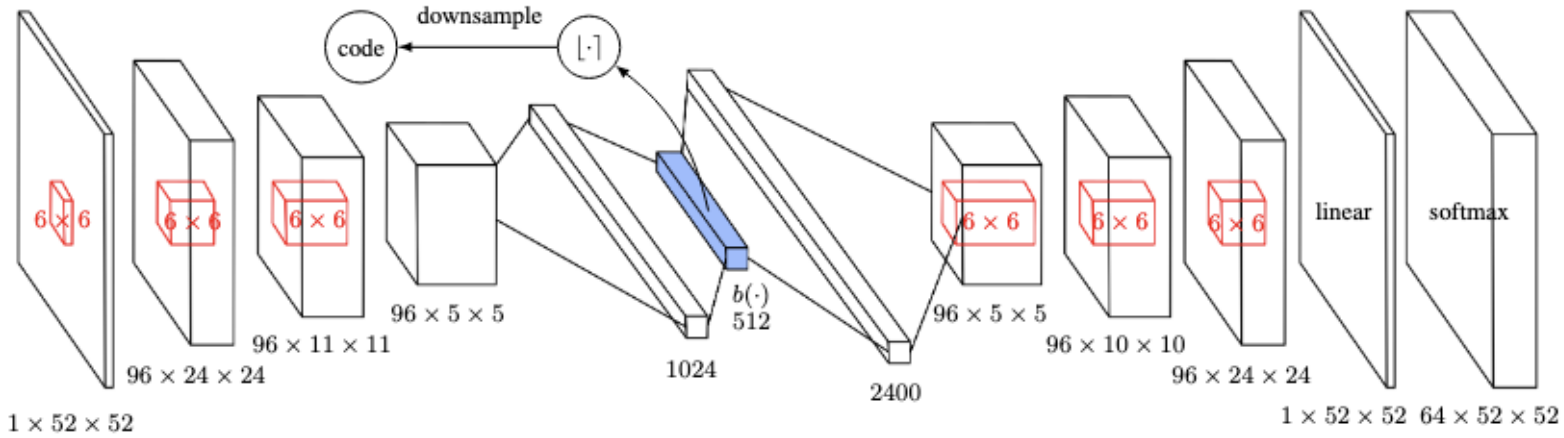

With bonus




MONTEZUMA'S REVENGE
FREEWAY
DQN
optimistic
$1/\sqrt{n}$
Score
7000
6000
5000
4000
3000
2000
1000
0
30
25
20
15
10
5
0
0
20
40
60
80
100
Training frames (millions)

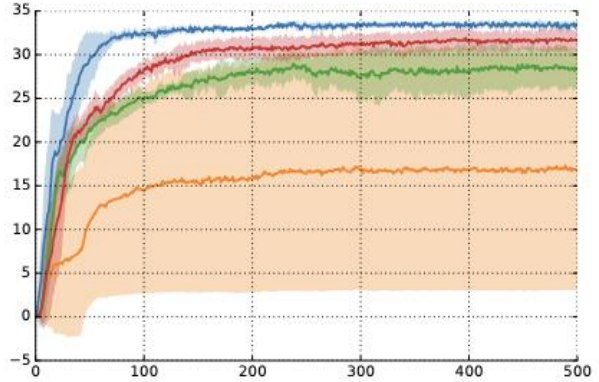
Idea : Counting with Hash Codes

- Learn to convert continuous inputs into discretized hash codes.
- Count the occurrences of the code
- How do we get the image encoding? E.g. using Autoencoders
  - ➢ Do similar states get the same hash? Maybe…

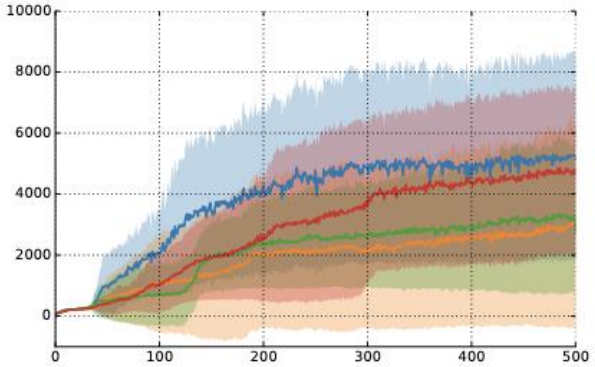


#Exploration: A Study of Count-Based Exploration for Deep Reinforcement Learning. Tang et al.

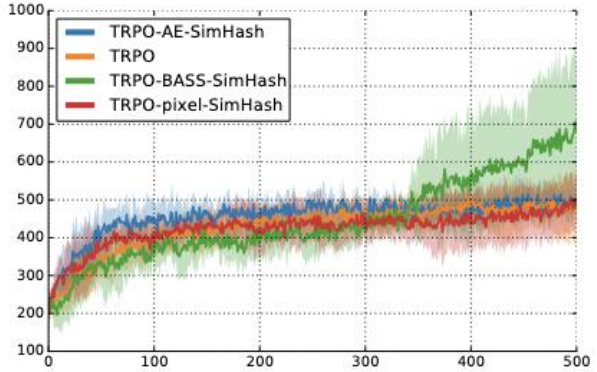
Does It Work?



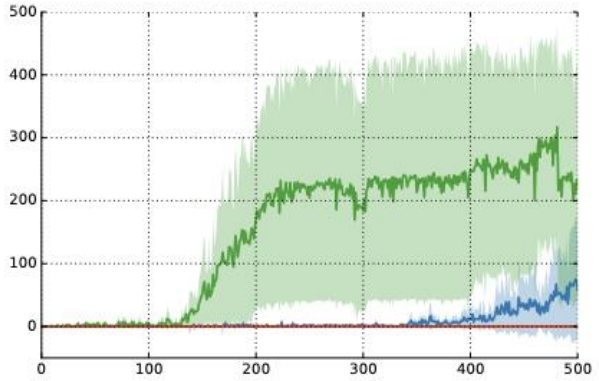
(a) Freeway



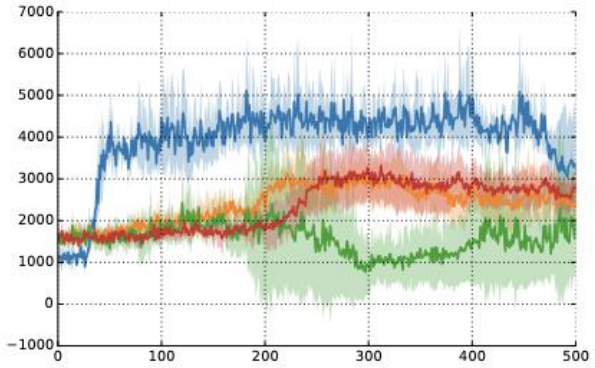
(b) Frostbite



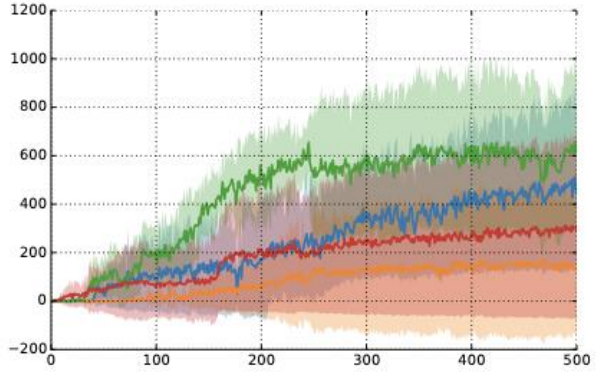
(c) Gravitar



(d) Montezuma's Revenge



(e) Solaris



(f) Venture

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Knowledge-based Exploration

- Idea: Maximize the knowledge about the world. The motivation is also known as *curiosity.* Knowledge could be in the form of:
  - Surprise
  - Unpredictability
  - How much learned about the world
  - Uncertainty

Computational Curiosity

- "The direct goal of curiosity and boredom is to improve the **world model**. The indirect goal is to ease the learning of new goal-directed action sequences."
- "The same complex mechanism which is used for 'normal' goal-directed learning is used for implementing curiosity and boredom. There is no need for devising a separate system which aims at improving the world model."
- "Curiosity Unit": reward is a function of the mismatch between model's current predictions and actuality. There is positive reinforcement whenever the system fails to correctly predict the environment.
- "Thus the usual credit assignment process ... encourages certain past actions in order to repeat situations similar to the mismatch situation." (planning to make your (internal) world model to fail)



Jurgen Schmidhuber, 1991, 1991, 1997

Slide credit K. Fragkiadaki

Computational Curiosity

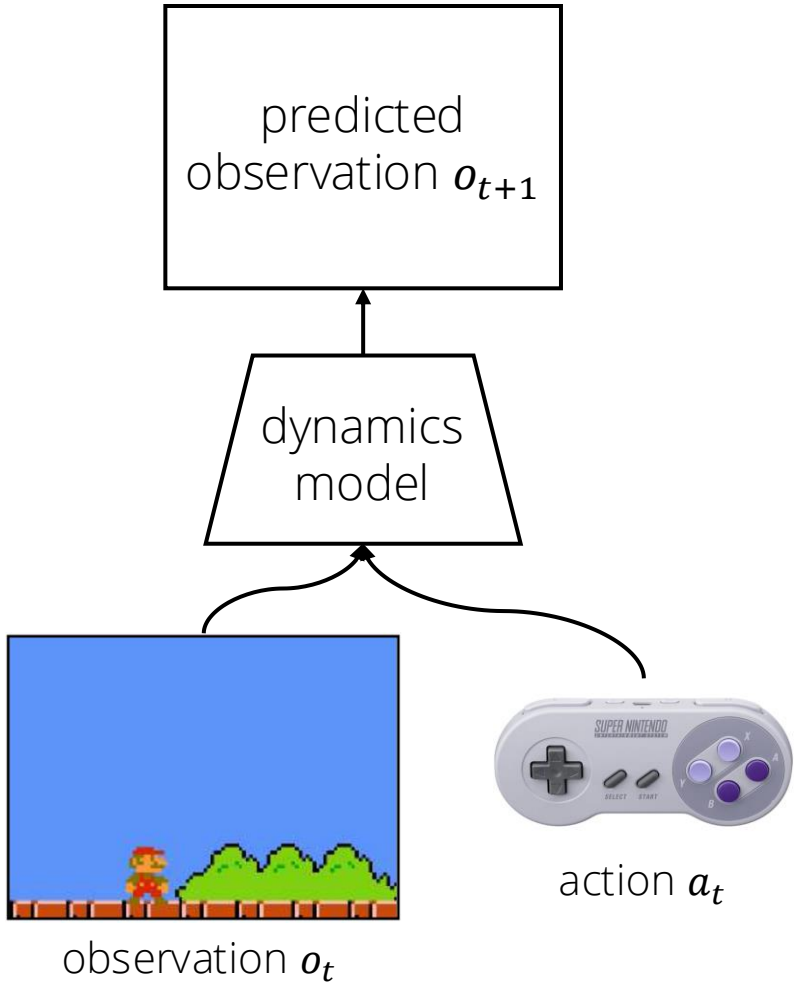


In other words:

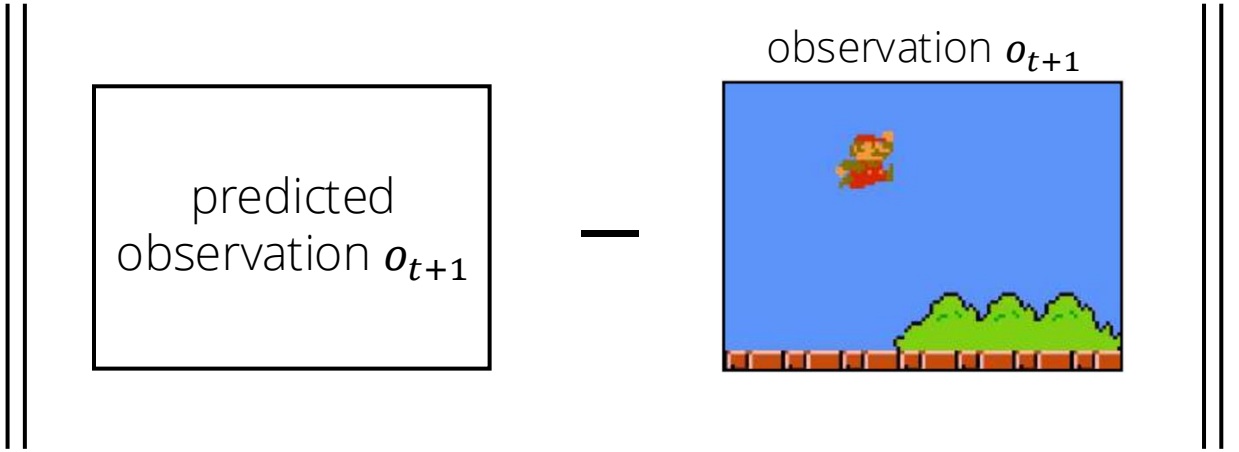
- Model learning and model improvement can be cast as the goals of goal-seeking behaviour.
- My goal is not to beat Atari but to improve my Atari model.
- OK. What is my reward then that trying to maximize that reward will lead to fast model learning?

Slide credit K. Fragkiadaki

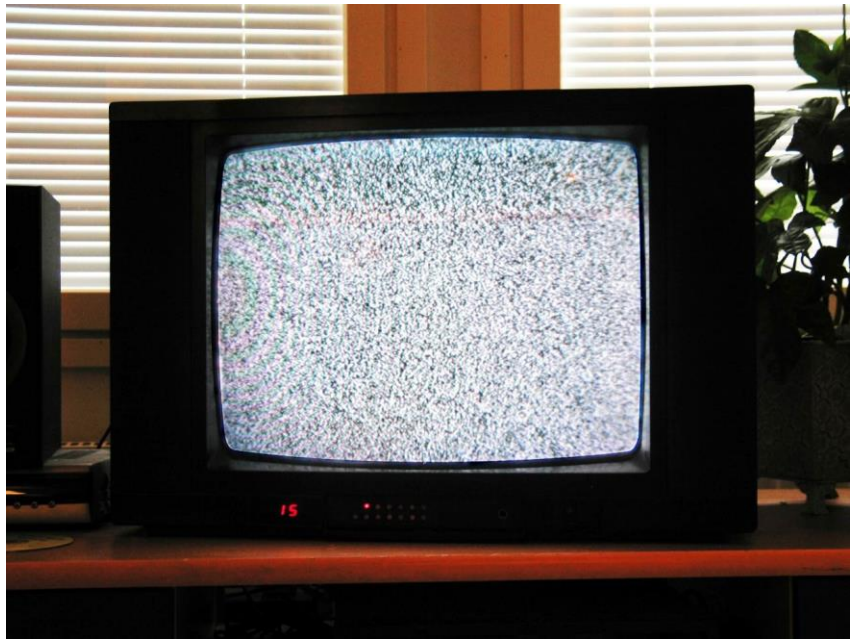
Curiosity Exploration



- Set the implicit reward $\mathcal{B}^t(s, a, s')$ as the prediction error of the dynamics model
- Action a is worth exploring if we are uncertain of its outcome

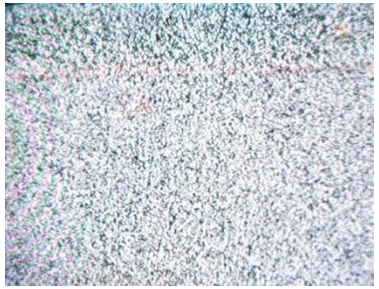


Is Pixel-based Prediction Error Desirable?

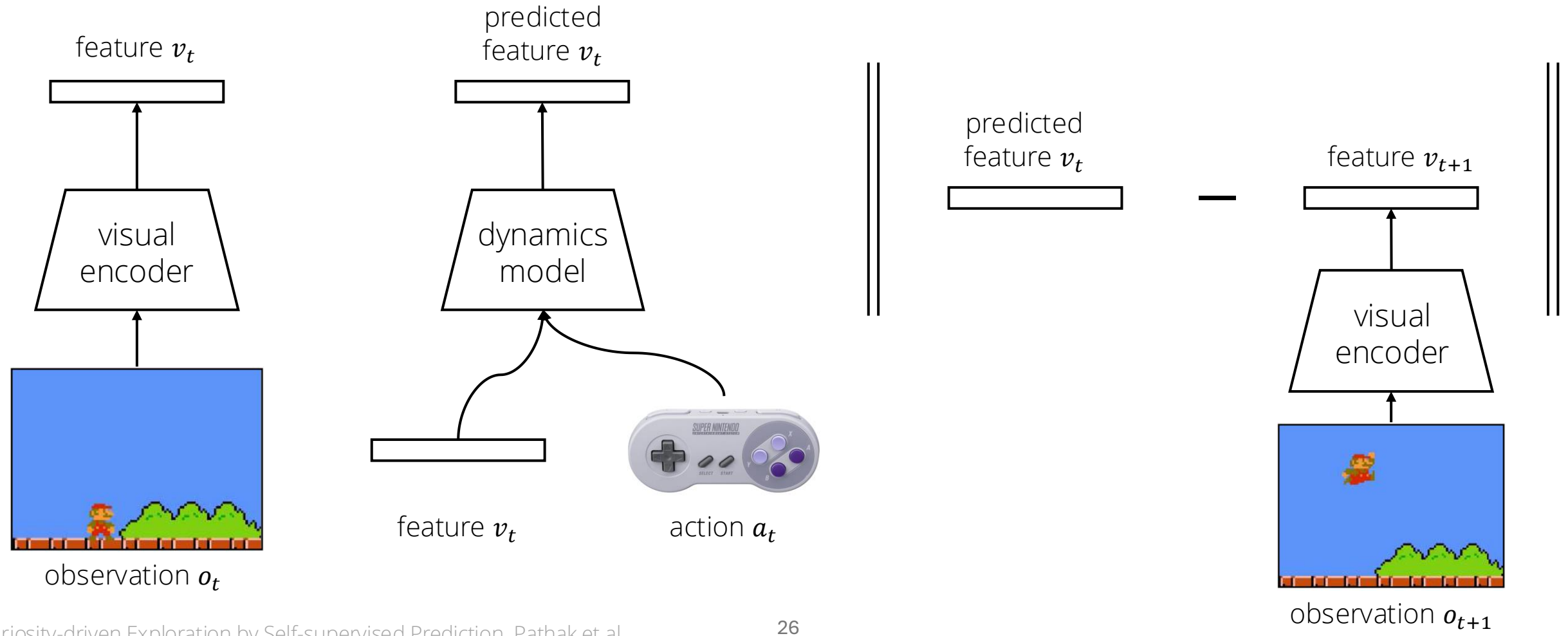


- Suppose the agent control to switch channels, and one channel if offline, showing white noise
- The prediction error on this channel will be very high
- The agent is encouraged to select this channel…

$$\left\| \text{predicted observation } \boldsymbol{o_{t+1}} - \text{observation } \boldsymbol{o_{t+1}} \right\|$$

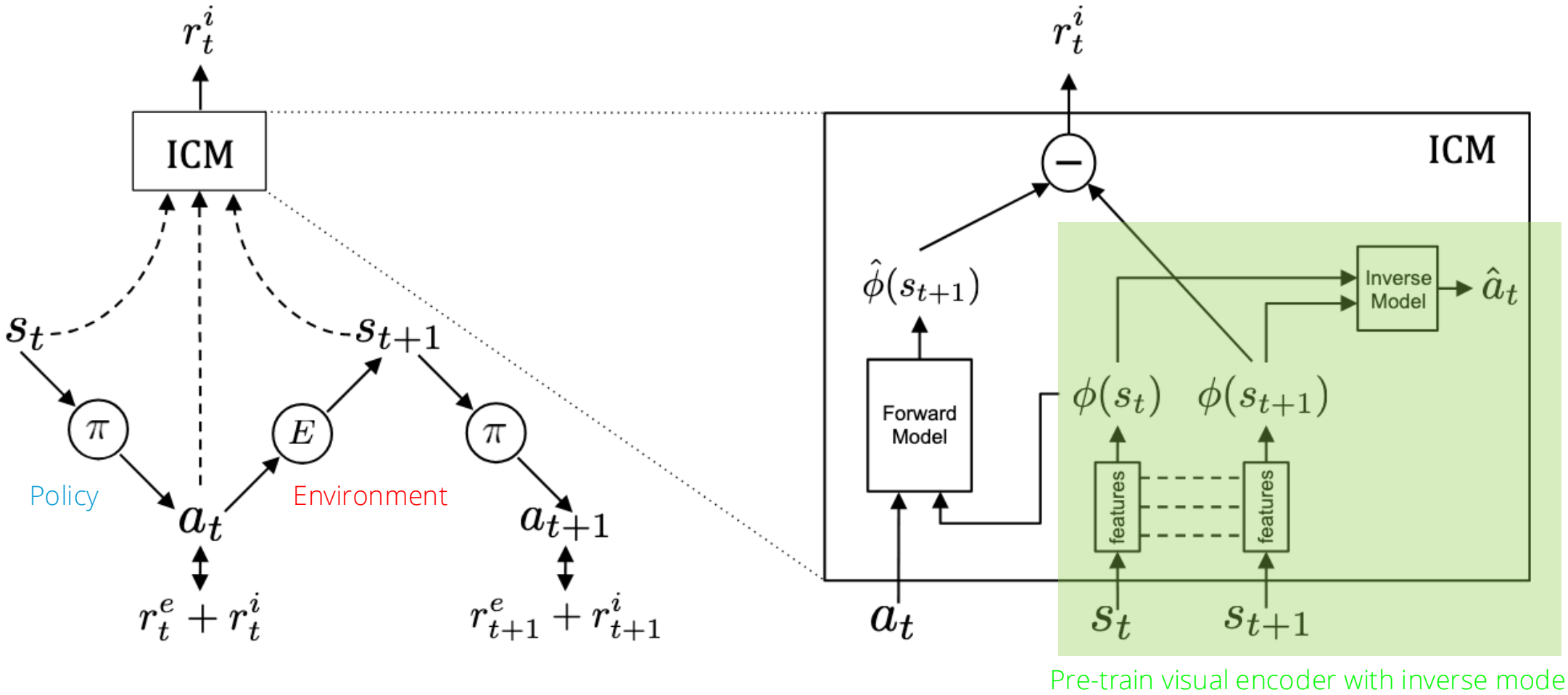


Idea: Curiosity as Prediction Errors in the Latent Feature Space



Curiosity-driven Exploration by Self-supervised Prediction. Pathak et al.

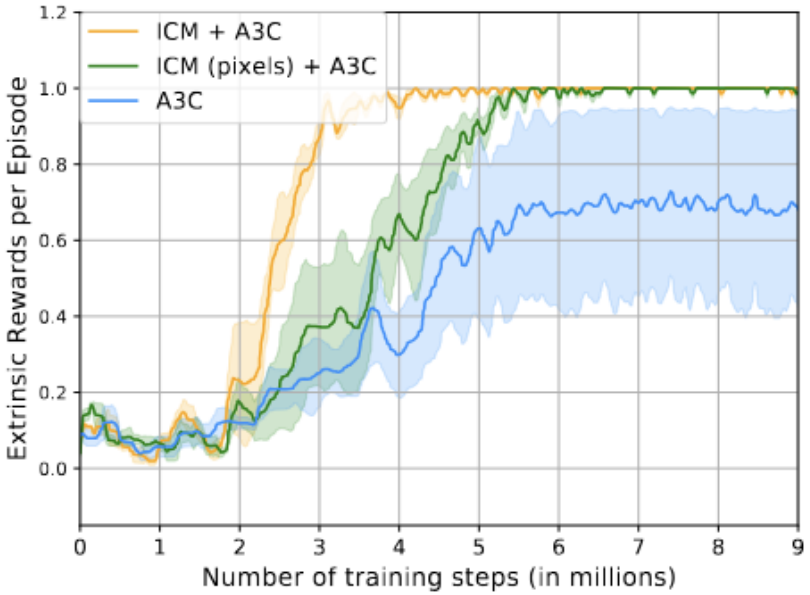
Regularize the Latent Feature Space: Only Encode Action-Related Information



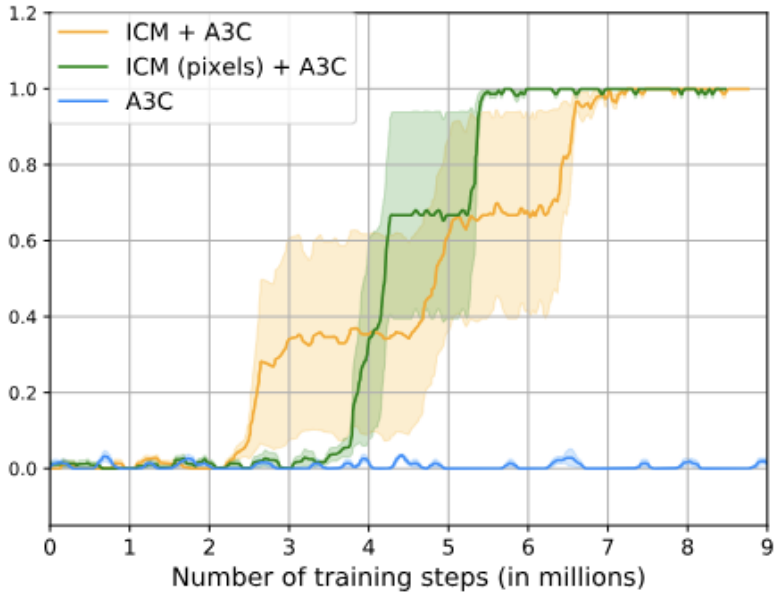
Pre-train visual encoder with inverse model

Curiosity-driven Exploration by Self-supervised Prediction. Pathak et al.

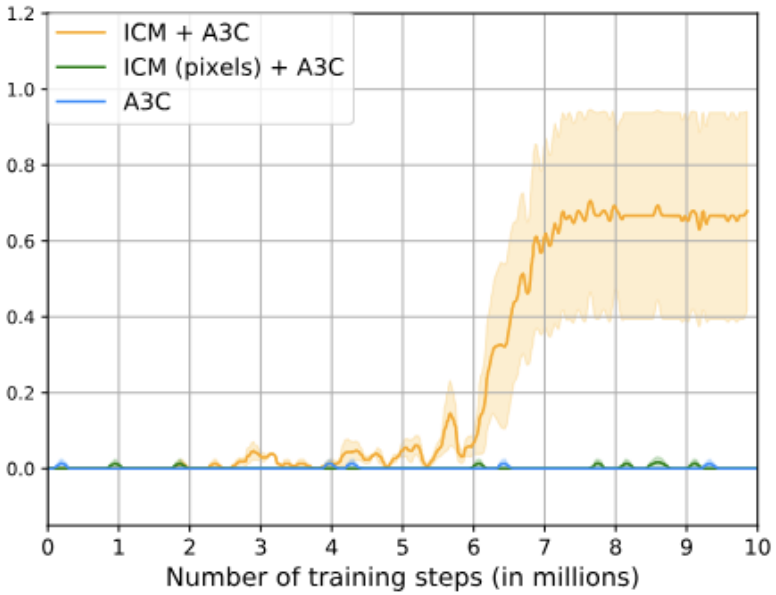
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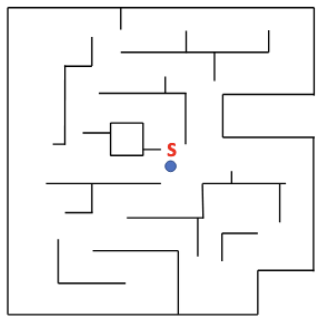
(a) "dense reward" setting



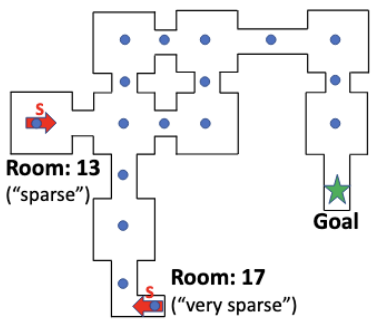
(b) "sparse reward" setting



(c) "very sparse reward" setting



(a) Train Map Scenario



(b) Test Map Scenario

Figure 4. Maps for VizDoom 3-D environment: (a) For generalization experiments (c.f. Section 4.3), map of the environment where agent is pre-trained only using curiosity signal without any reward from environment. 'S' denotes the starting position. (b) Testing map for VizDoom experiments. Green star denotes goal location. Blue dots refer to 17 agent spawning locations in the map in the "dense" case. Rooms 13, 17 are the fixed start locations of agent in "sparse" and "very sparse" reward cases respectively. Note that textures are also different in train and test maps.

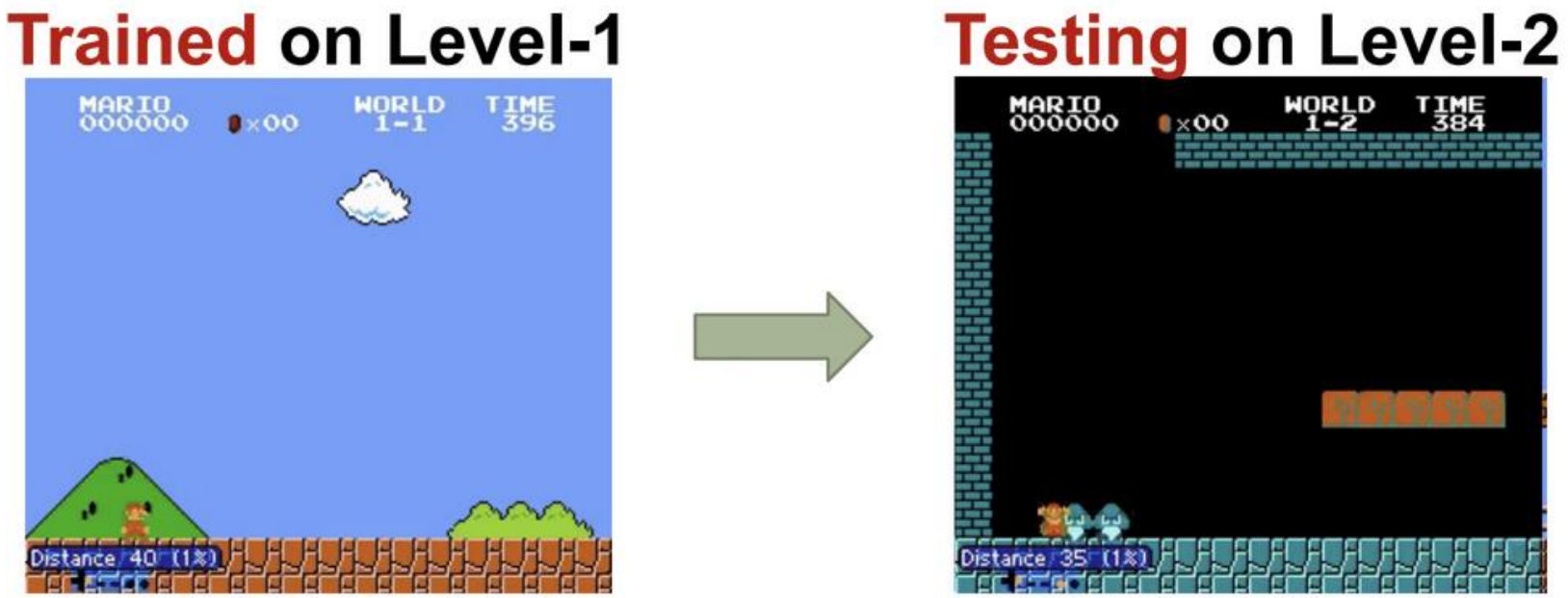
RL Objective:

$$R^t(s, a, s') = r(s, a, s') + \mathcal{B}^t(s, a, s')$$

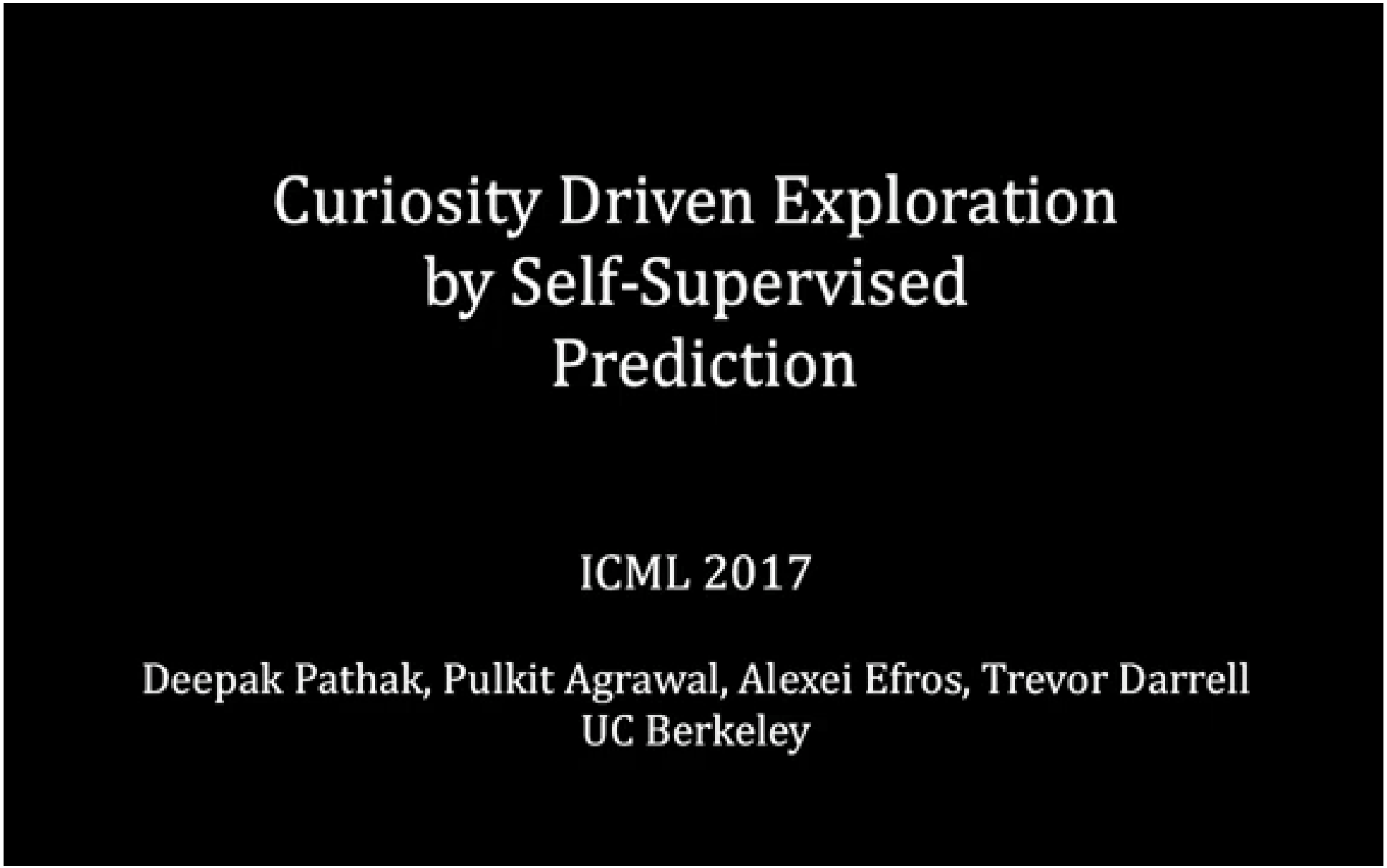
What if We Train the Policy with Only Intrinsic Reward?

- RL Objective:

$$R^t(s,a,s') = \cancel{r(s,a,s')} + \mathcal{B}^t(s,a,s')$$



Curiosity-driven Exploration by Self-supervised Prediction. Pathak et al.



Curiosity-driven Exploration by Self-supervised Prediction. Pathak et al.

What Could be Wrong?

- The source of high predictions errors result from:
  1. Amount of training data (epistemic uncertainty): few similar examples were seen by the predictor
  2. Stochasticity (aleatoric uncertainty): stochastic transitions are a source of such error for forward dynamics prediction
  3. Model misspecification: necessary information is missing, or the model class is too limited to fit the complexity of the dynamic function
  4. Learning dynamics: the optimization process fails to find a predictor in the model class that best approximates the dynamic function

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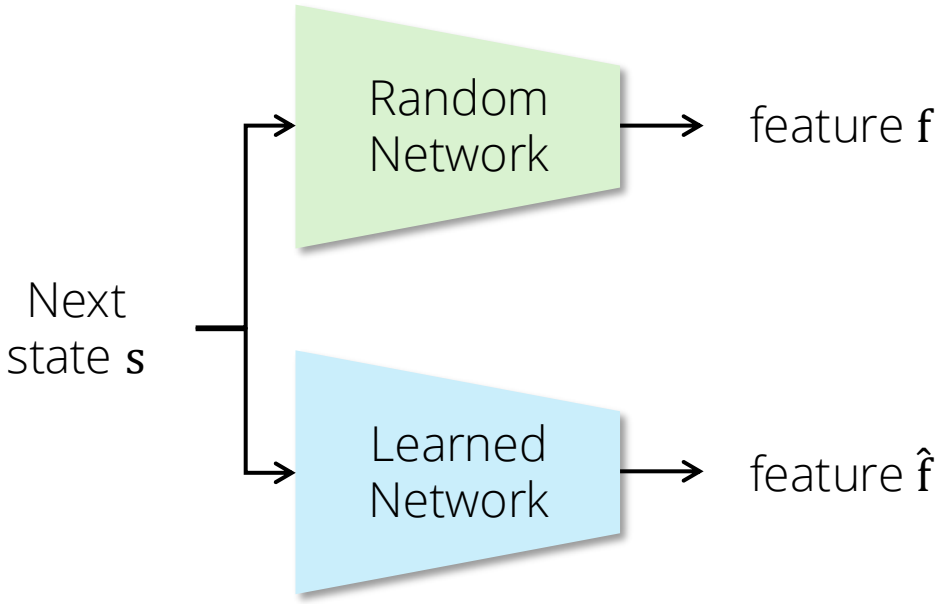
Can we fix them?

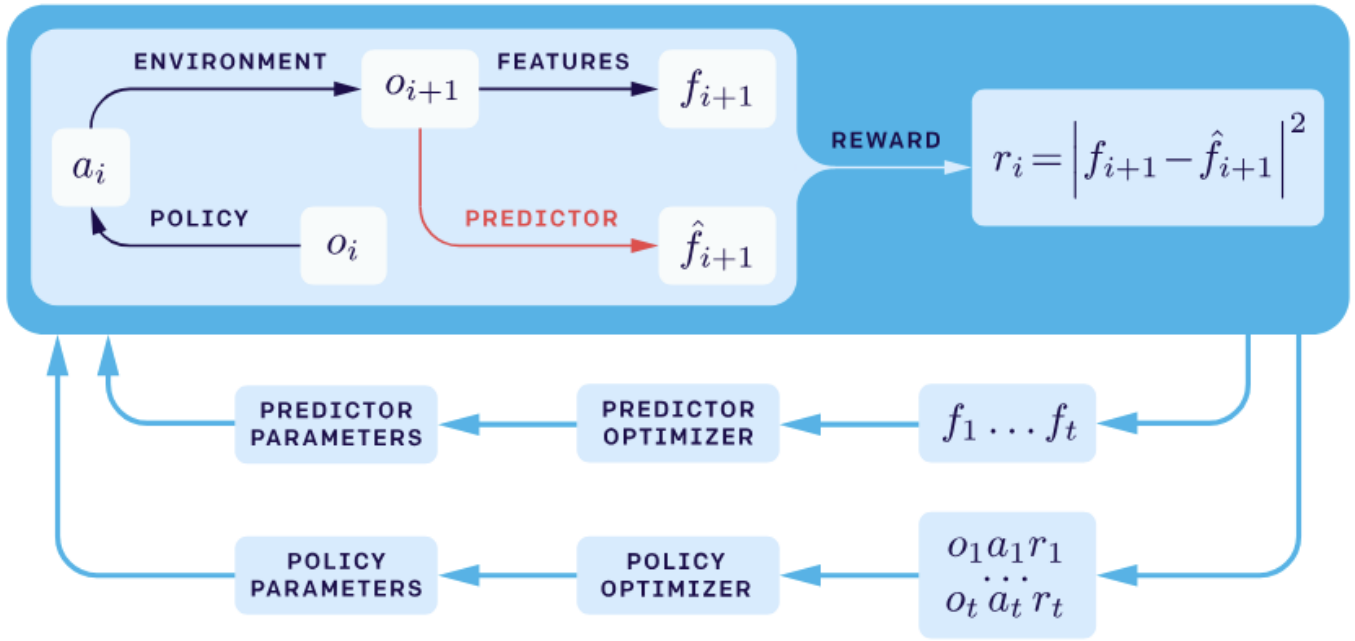
✗ 2. **Stochasticity** (aleatoric uncertainty): stochastic transitions are a source of such error for forward dynamics prediction

✗ 3. **Model misspecification**: necessary information is missing, or the model class is too limited to fit the complexity of the dynamic function

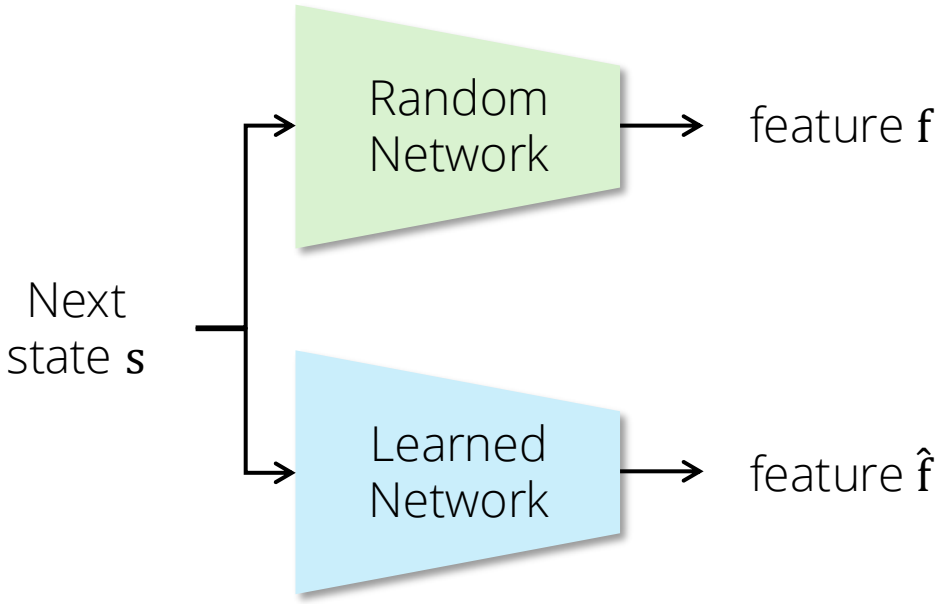
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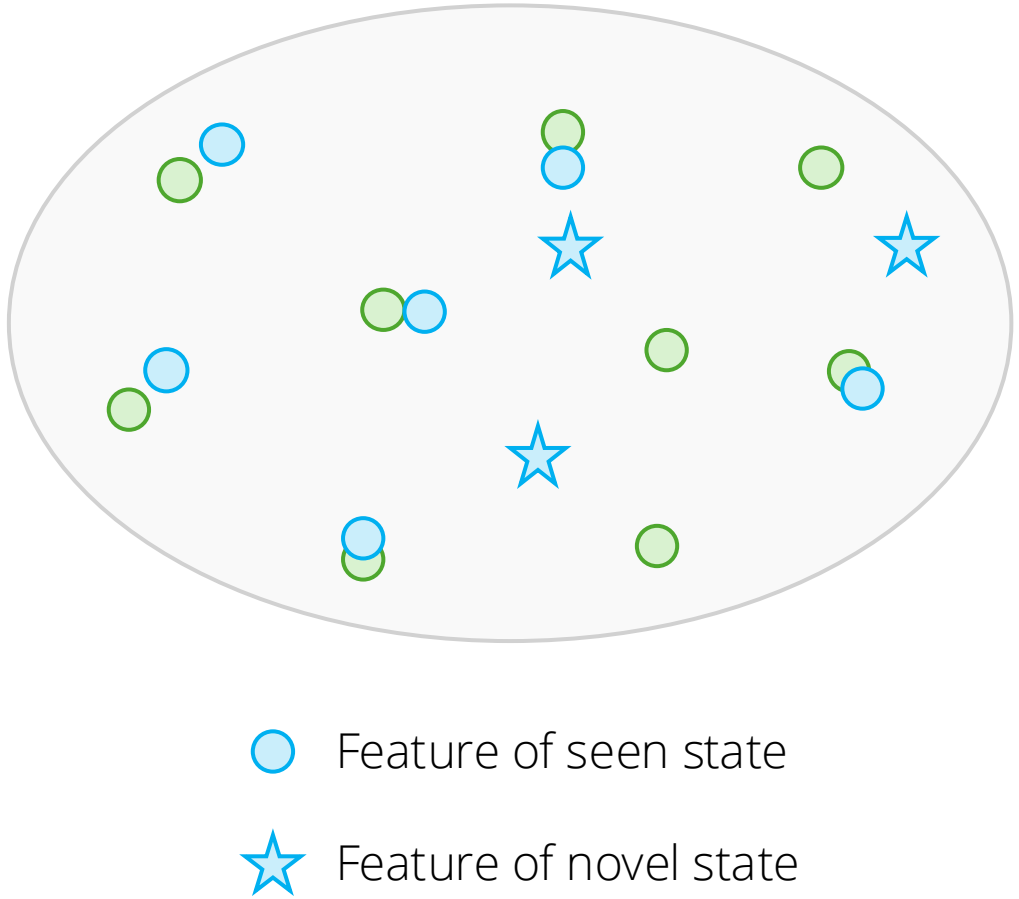
Exploration By Random Network Distillation





Novel States Result in Larger Feature Distance





What Could be Wrong?

Solution: (1) Bypass learning forward dynamics model but deterministically learn to encode future states. (2) Smoothness prior in neural network enhance the robustness of novel-state estimation

✕ 2. **Stochasticity** (aleatoric uncertainty): stochastic transitions are a source of such error for forward dynamics prediction

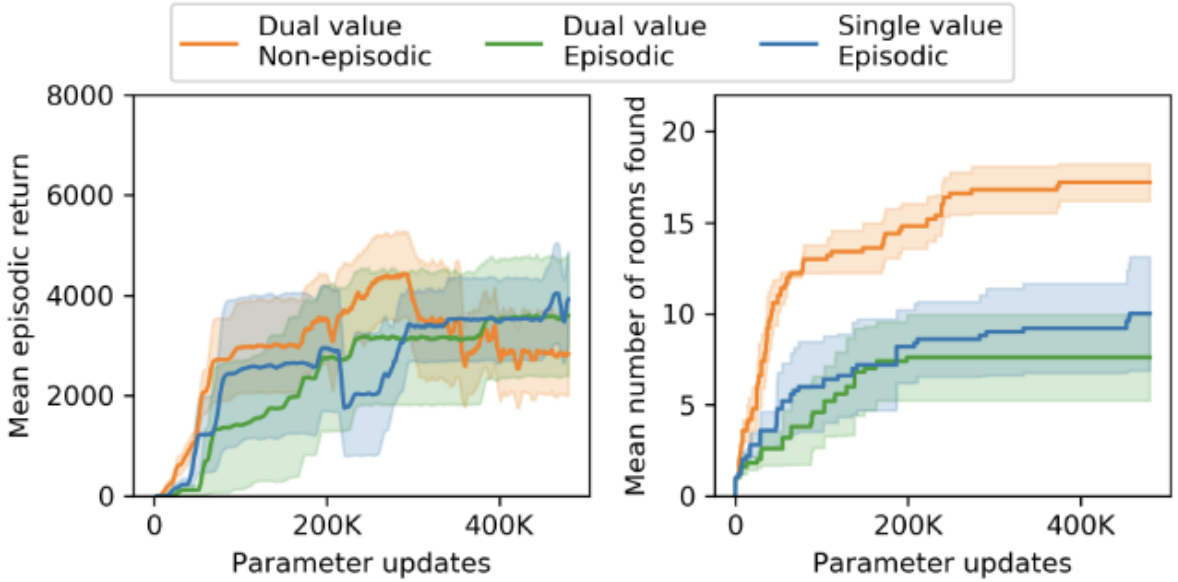
✕ 3. **Model misspecification**: necessary information is missing, or the model class is too limited to fit the complexity of the dynamic function

Solution: Build the target network and the predictor with similar model class

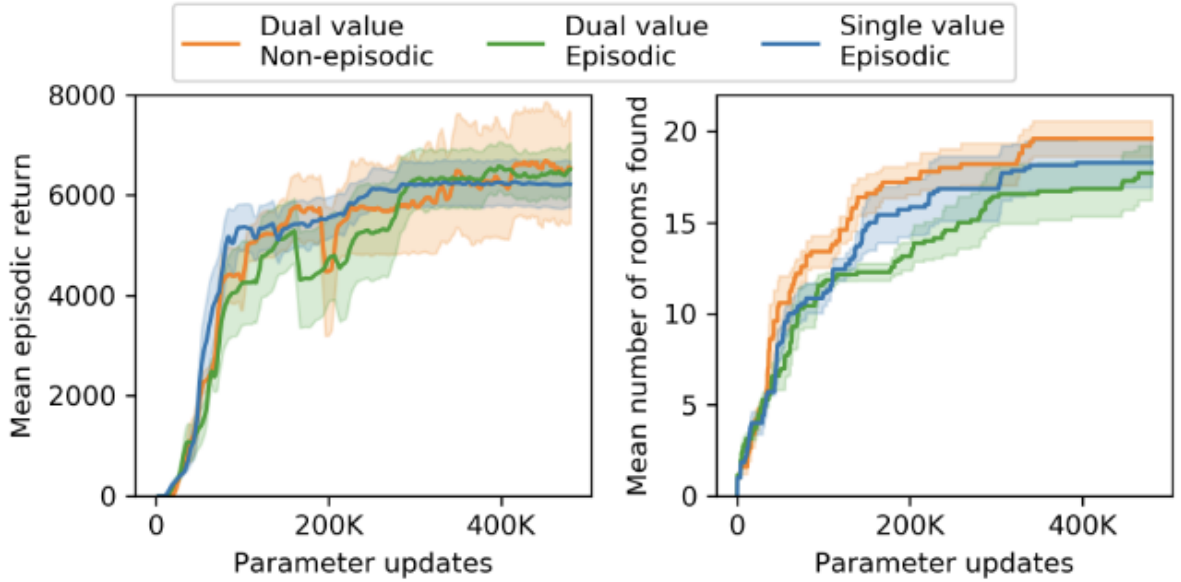
Details Matter: Separate Return Streams

- For extrinsic rewards: episodic return streams and an independent value head $\mathbf{V_E}$
- For extrinsic rewards: non-episodic return streams and an independent value head $\mathbf{V_I}$
- Train the agent with PPO using $\mathbf{V} = \mathbf{V_E} + \mathbf{V_I}$

Non-episodic Intrinsic Return Streams Matter

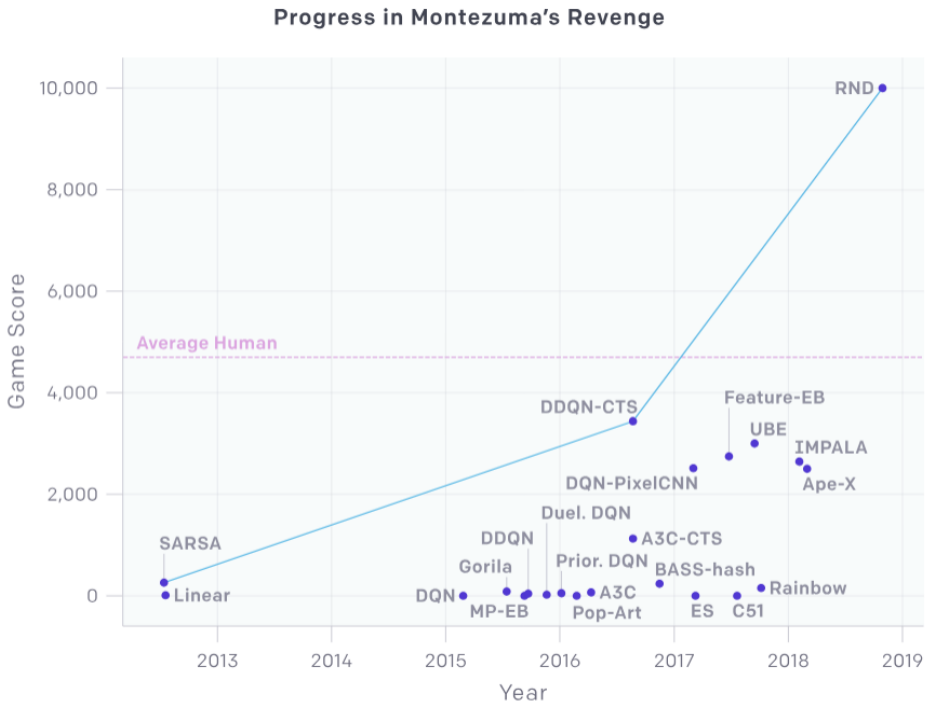


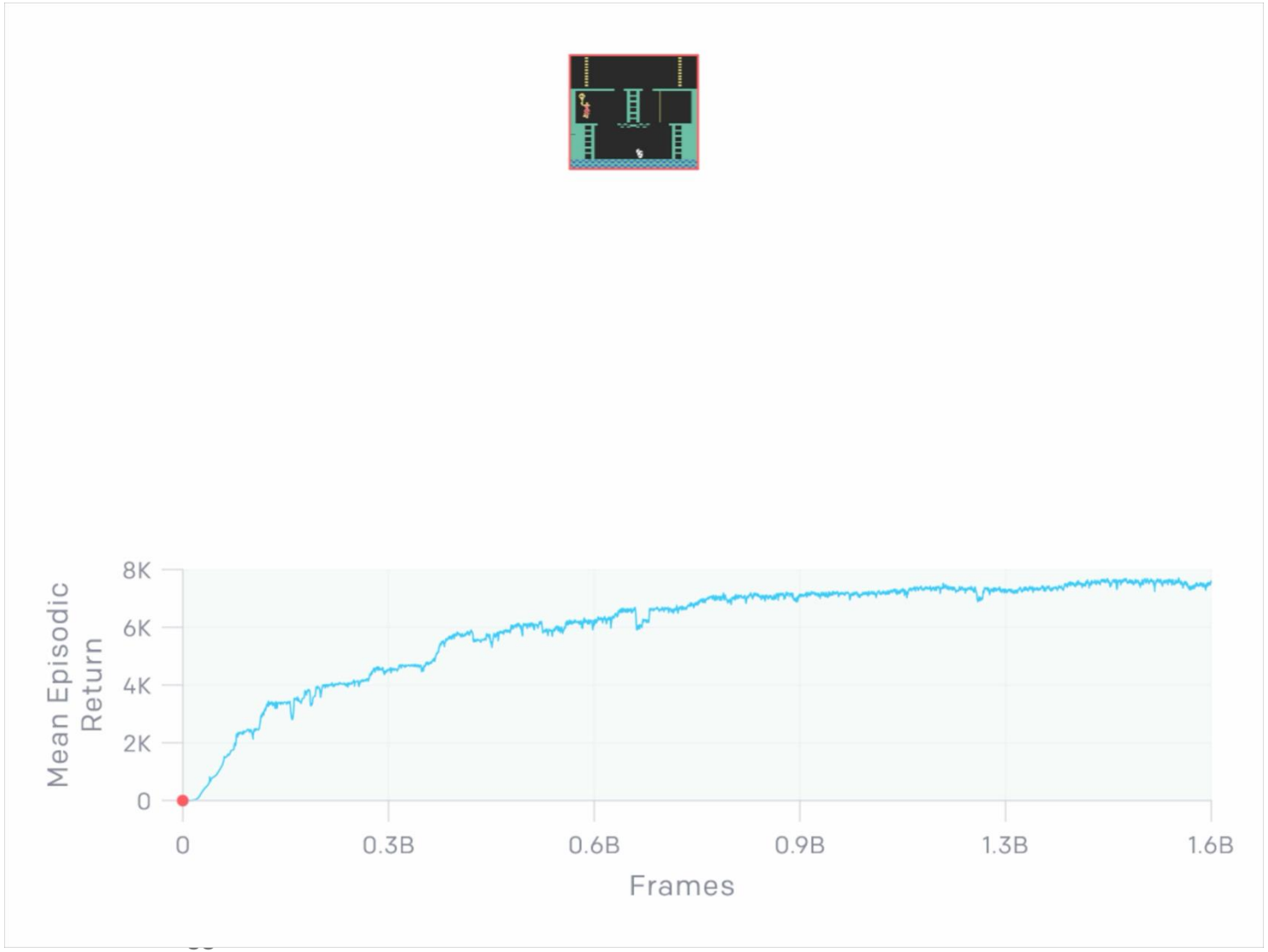
(a) RNN policies



(b) CNN policies

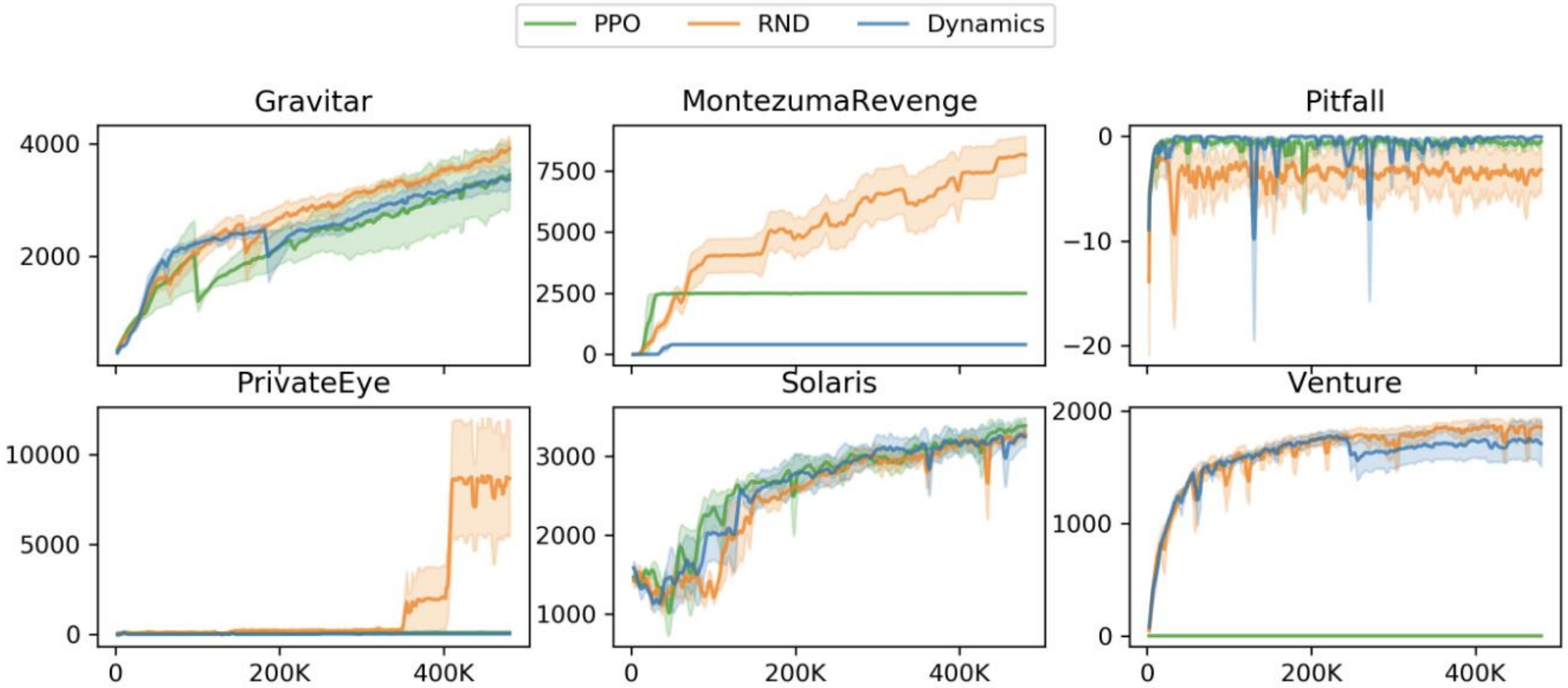
State-of-the-Art in Montezuma's Revenge





Exploration by Random Network Distillation. Burda et al.

Outperforms PPO



Exploration by Random Network Distillation. Burda et al.

How to Model Exploration?

- We can re-formulate the objective of reinforcement learning:

$$R^t(s, a, s') = r(s, a, s') + \mathcal{B}^t(s, a, s')$$

extrinsic reward (related to the task)

intrinsic reward (unrelated to the task)

- Types of intrinsic rewards:
  1. Data-based exploration (Optimistic exploration)
  2. Knowledge-based exploration
  3. Skill-based exploration
  4. Reachability exploration
  5. Posterior sampling for exploration

Empowerment and Mutual Information

- The concept of *empowerment*: an attempt at formalizing and quantifying the degrees of freedom (or options) that an organism or agent has as a proxy for "*preparedness*"

- *Preparedness* can be seen as:
  - Keeping options open
  - Influence over the environment
  - Resilience to uncertainty

- Empowerment through Mutual Information Maximization: Find policy $\pi(a|y,x)$ that maximizes mutual information $\mathcal{I}(x;y)$

- Mutual information:

$$\mathcal{I}(x;y) = \mathcal{H}(x) - H(x|y)$$

where $\mathcal{H}$ denotes entropy (the higher entropy the higher uncertainty is)

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- Mutual information:

How much information you know about x when you perform action y

$$\mathcal{I}(x;y) = \mathcal{H}(x) - H(x|y)$$

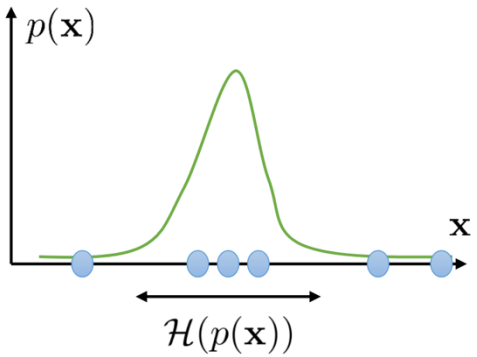
where $\mathcal{H}$ denotes entropy (the higher entropy the higher uncertainty is)

Some Useful Identities

$p(\mathbf{x})$ distribution (e.g., over observations $\mathbf{x}$)

$$\mathcal{H}(p(\mathbf{x})) = -E_{\mathbf{x}\sim p(\mathbf{x})}[\log p(\mathbf{x})]$$

entropy – how "broad" $p(\mathbf{x})$ is



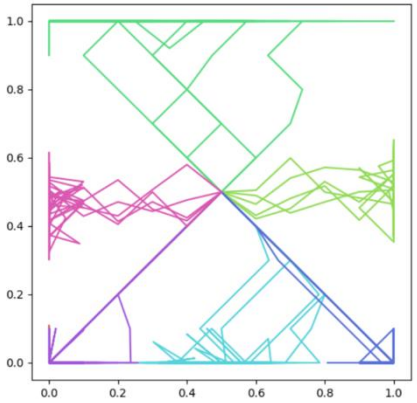
$$\mathcal{I}(\mathbf{x};\mathbf{y}) = D_{\mathrm{KL}}(p(\mathbf{x},\mathbf{y})\|p(\mathbf{x})p(\mathbf{y}))$$

$$= E_{(\mathbf{x},\mathbf{y})\sim p(\mathbf{x},\mathbf{y})}\left[\log \frac{p(\mathbf{x},\mathbf{y})}{p(\mathbf{x})p(\mathbf{y})}\right]$$

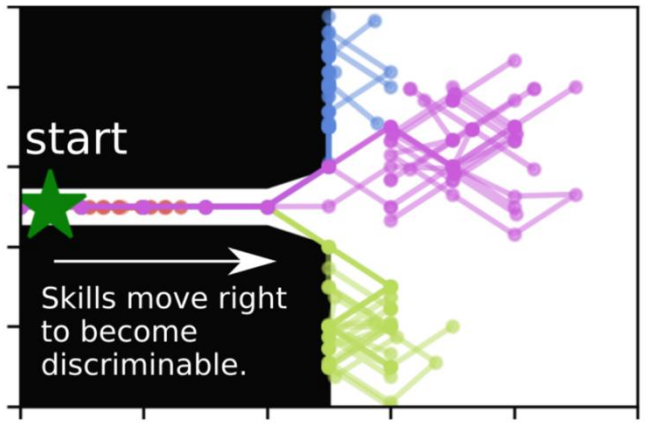
$$= \mathcal{H}(p(\mathbf{y})) - \mathcal{H}(p(\mathbf{y}|\mathbf{x}))$$

Slide credit S. Levine

Skill-based Exploration: Diversity is All You Need (DAIYN)



(a) 2D Navigation

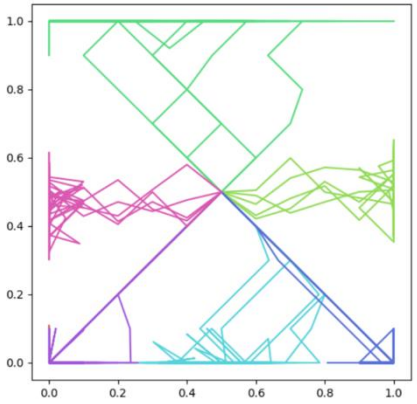


(b) Overlapping Skills

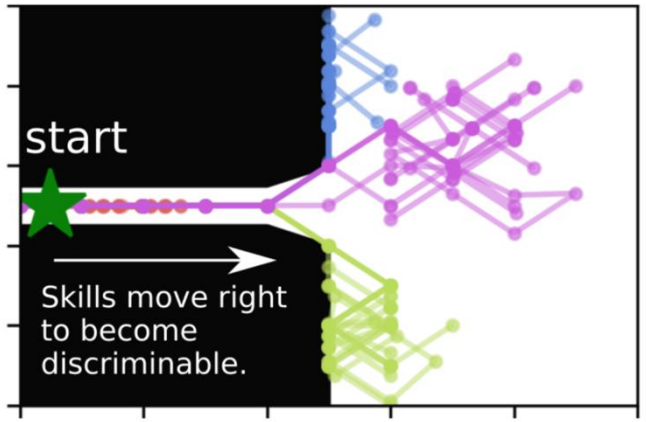
- This paper defines *skill* as the states that the agent visits
- The core idea is to find *skill* that induces actions to visit different states. These states should be distinguishable
- How to formulate the problem?

Diversity is All You Need: Learning Skills without a Reward Function .Eysenbach et al.

Skill-based Exploration: Diversity is All You Need (DAIYN)



(a) 2D Navigation



(b) Overlapping Skills

- Let $q_\phi(z|s)$ be the discriminator probability of skill z given state s parameterized by ϕ.
- Intrinsic reward as the discriminability of skill z given state s:

$$r_z(s,a) \coloneqq \log q_\phi(z|s) - \log p(z)$$

- The objective of the policy:

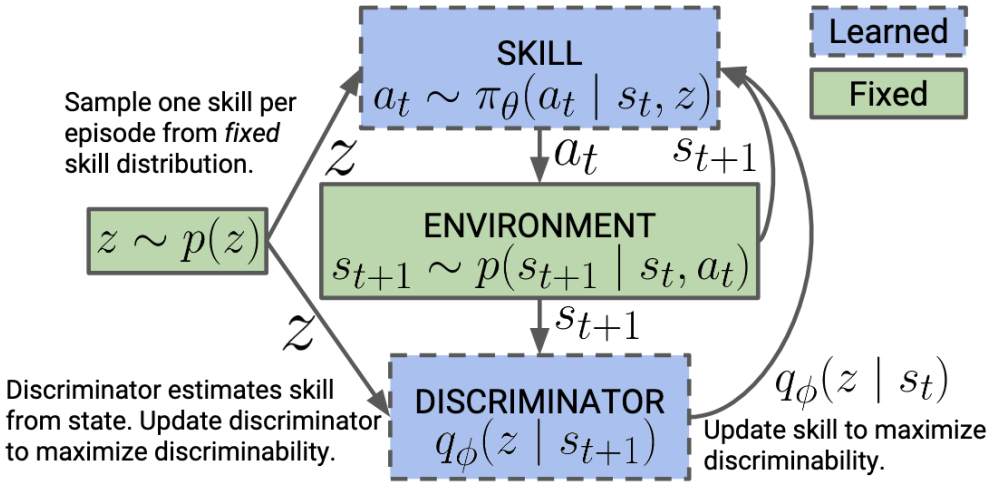
$$\pi_\theta(a|s,z) = \operatorname*{argmax}_\pi \sum_z \mathbb{E}_{s\sim\pi(s|z)}[r(s,z)]$$

reward states that are unlikely for other $z' \neq z$

Slide credit S. Levine

Diversity is All You Need: Learning Skills without a Reward Function .Eysenbach et al.

Skill-based Exploration: Diversity is All You Need (DAIYN)



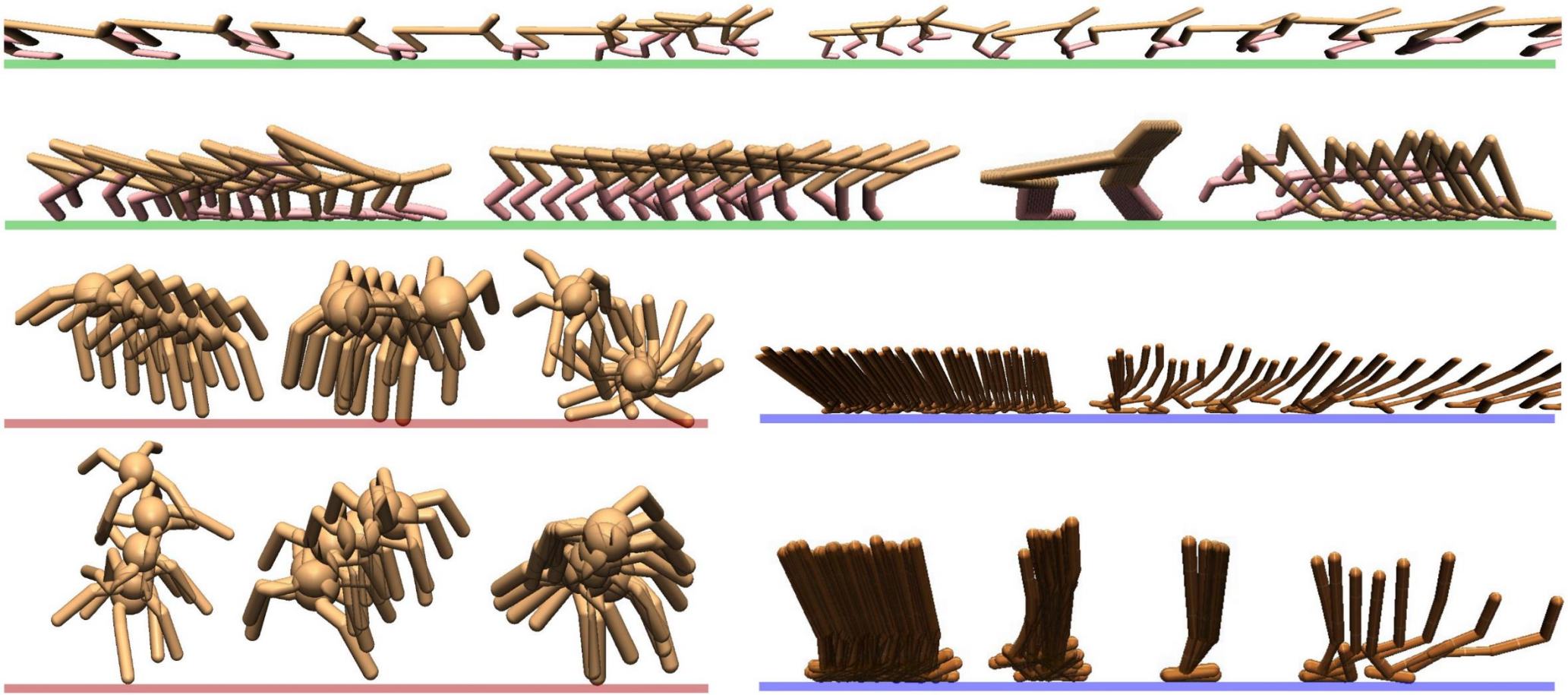
$p(z)$ is fixed so that model won't collapse

Algorithm 1: DIAYN

while *not converged* **do**
- Sample skill $z \sim p(z)$ and initial state $s_0 \sim p_0(s)$
- **for** $t \leftarrow 1$ **to** *steps_per_episode* **do**
  - Sample action $a_t \sim \pi_\theta(a_t \mid s_t, z)$ from skill.
  - Step environment: $s_{t+1} \sim p(s_{t+1} \mid s_t, a_t)$.
  - Compute $q_\phi(z \mid s_{t+1})$ with discriminator.
  - Set skill reward $r_t = \log q_\phi(z \mid s_{t+1}) - \log p(z)$
  - Update policy (θ) to maximize r_t with SAC.
  - Update discriminator (ϕ) with SGD.

Diversity is All You Need: Learning Skills without a Reward Function .Eysenbach et al.

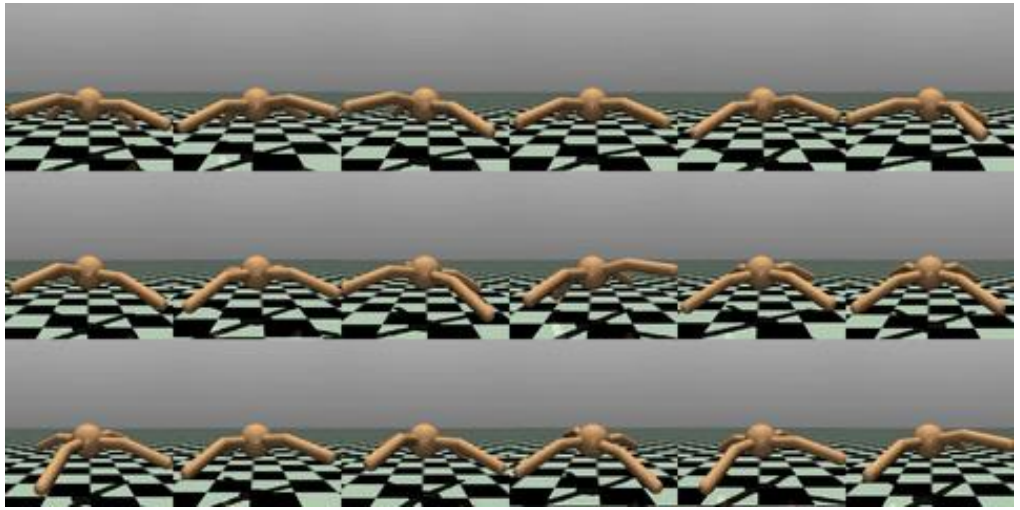
Example of Learned Skills

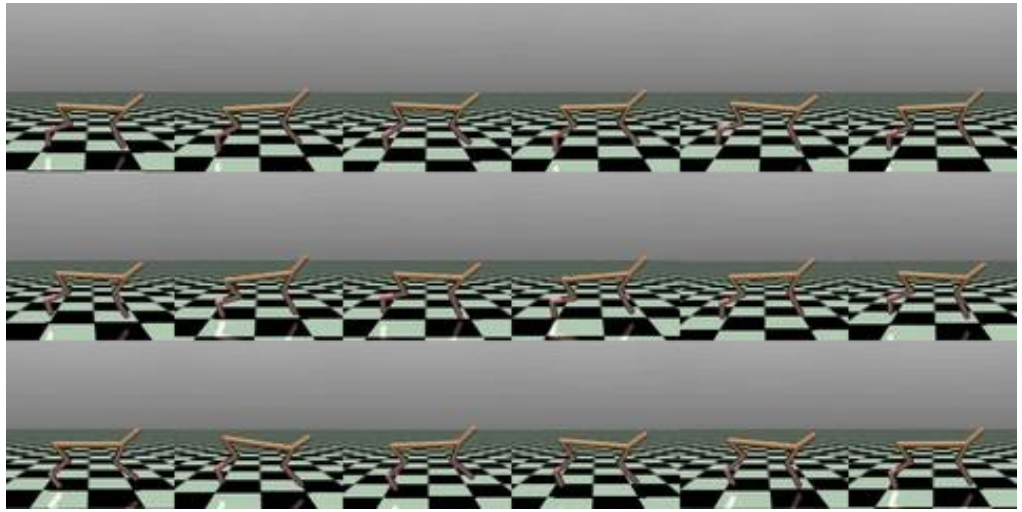


Slide credit S. Levine

Diversity is All You Need: Learning Skills without a Reward Function .Eysenbach et al.

Issues: Discovered Skills are not Useful / Repetitive









Unsupervised skill discovery is underexplored!

Can a MISL Fly? Analysis and Ingredients for Mutual Information Skill Learning. Zheng et al

The Connection to Mutual Information

$$\pi(\mathbf{a}|\mathbf{s}, z) = \arg\max_{\pi} \sum_{z} E_{\mathbf{s}\sim\pi(\mathbf{s}|z)}[r(\mathbf{s}, z)]$$

$$r(\mathbf{s}, z) = \log p(z|\mathbf{s})$$

$$I(z, \mathbf{s}) = H(z) - H(z|s)$$

maximized by using uniform prior $p(z)$

minimized by maximizing $\log p(z|\mathbf{s})$

Slide credit S. Levine

Diversity is All You Need: Learning Skills without a Reward Function .Eysenbach et al.

How to Model Exploration?

- We can re-formulate the objective of reinforcement learning:

$$R^t(s, a, s') = r(s, a, s') + \mathcal{B}^t(s, a, s')$$

extrinsic reward (related to the task)

intrinsic reward (unrelated to the task)

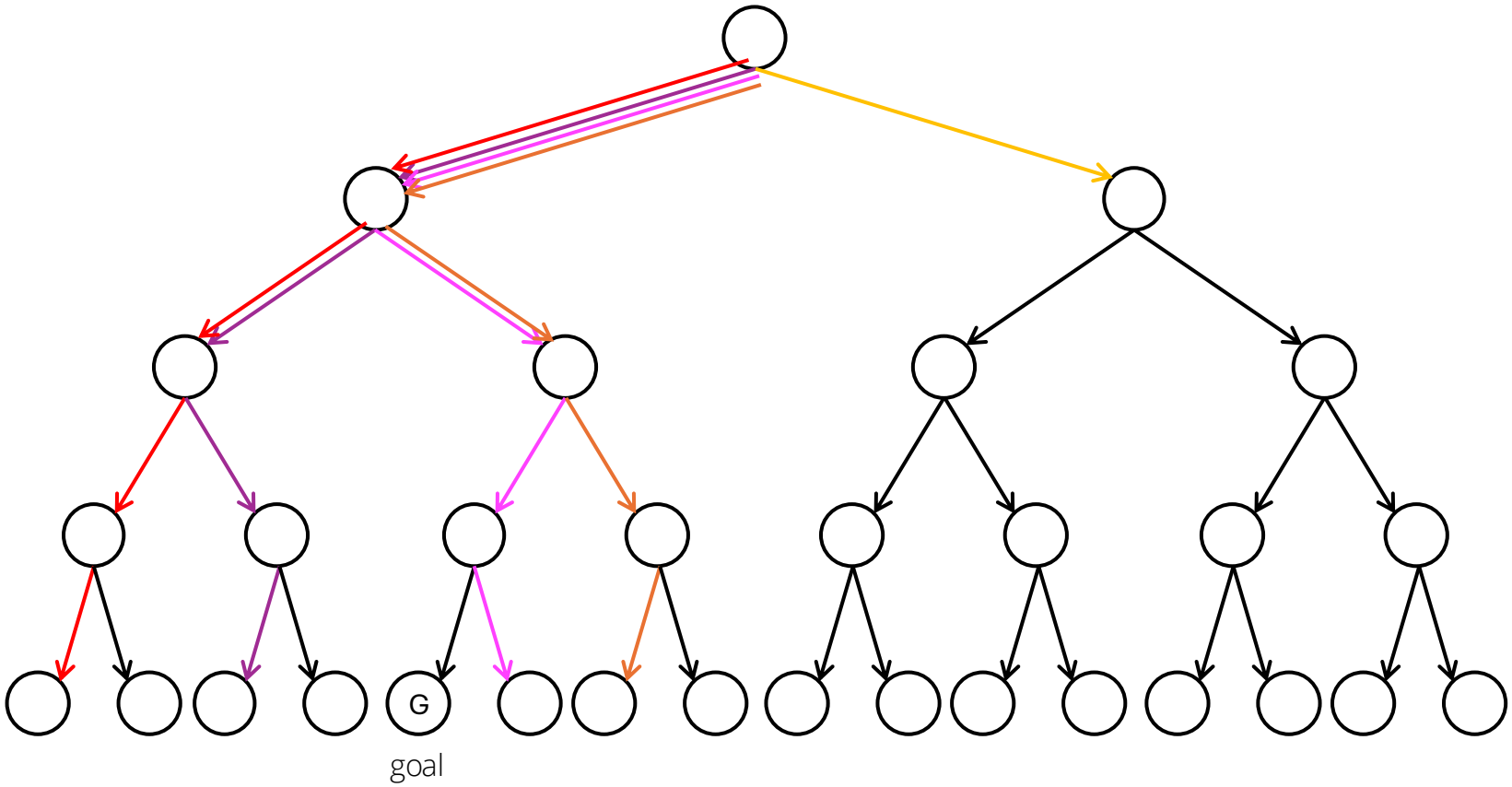
- Types of intrinsic rewards:
  1. Data-based exploration (Optimistic exploration)
  2. Knowledge-based exploration
  3. Skill-based exploration
  4. Reachability exploration
  5. Posterior sampling for exploration

Exploration is Hard…

- Hard exploration problem:
  - When the reward is sparse, precise sequences of many (e.g. hundreds or more) actions must be taken between obtaining rewards
  - When the reward is deceptive, it actually provides misleading feedback for reaching the overall global objective, which can lead to getting stuck on local optima

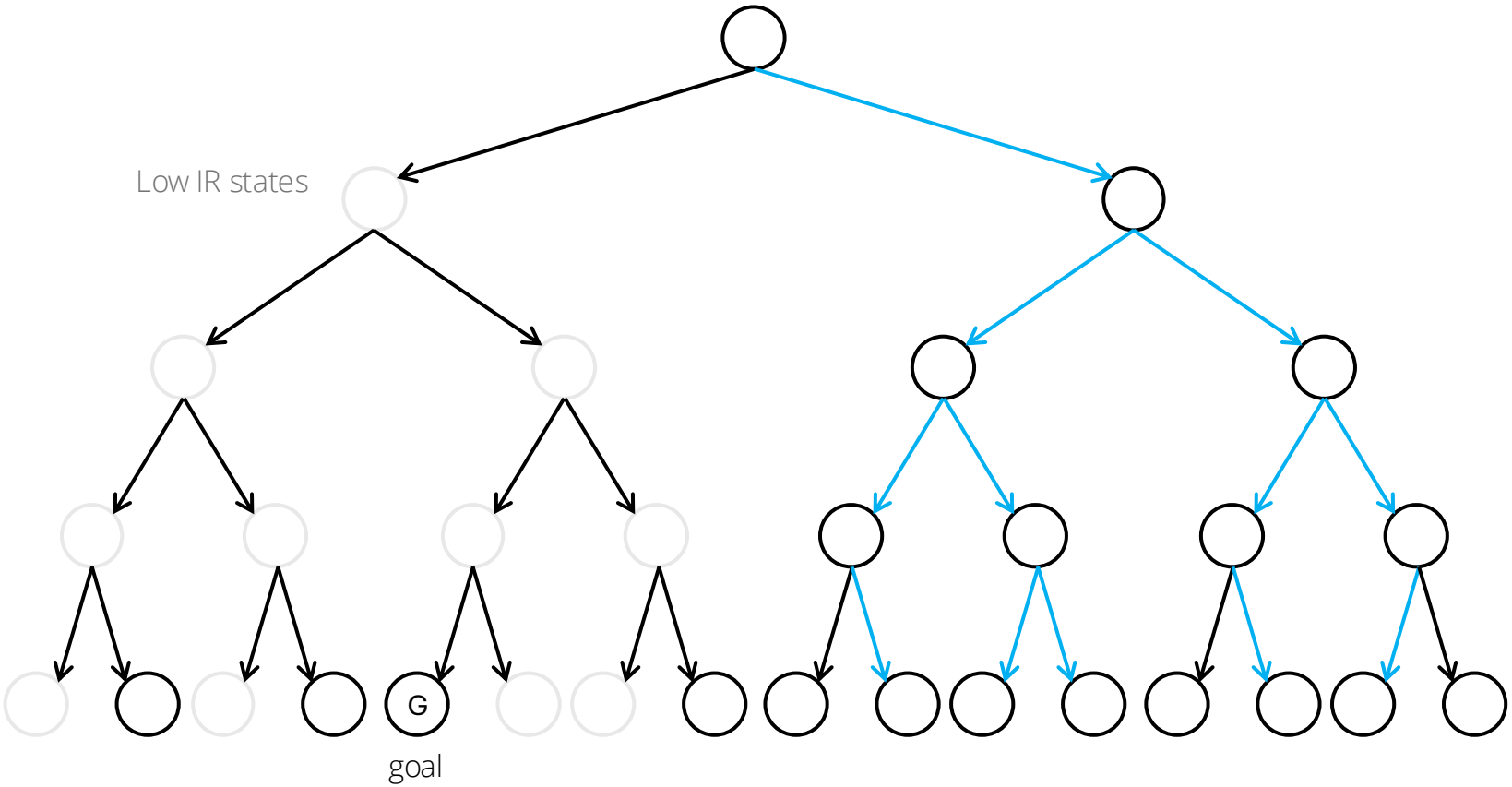
- Exploration methods based intrinsic motivations have two problems:
  - Detachment: an agent become detached from the frontiers of high intrinsic reward
  - Derailment: an agent derail from the discovered high-reward state

Previous Exploration Strategy



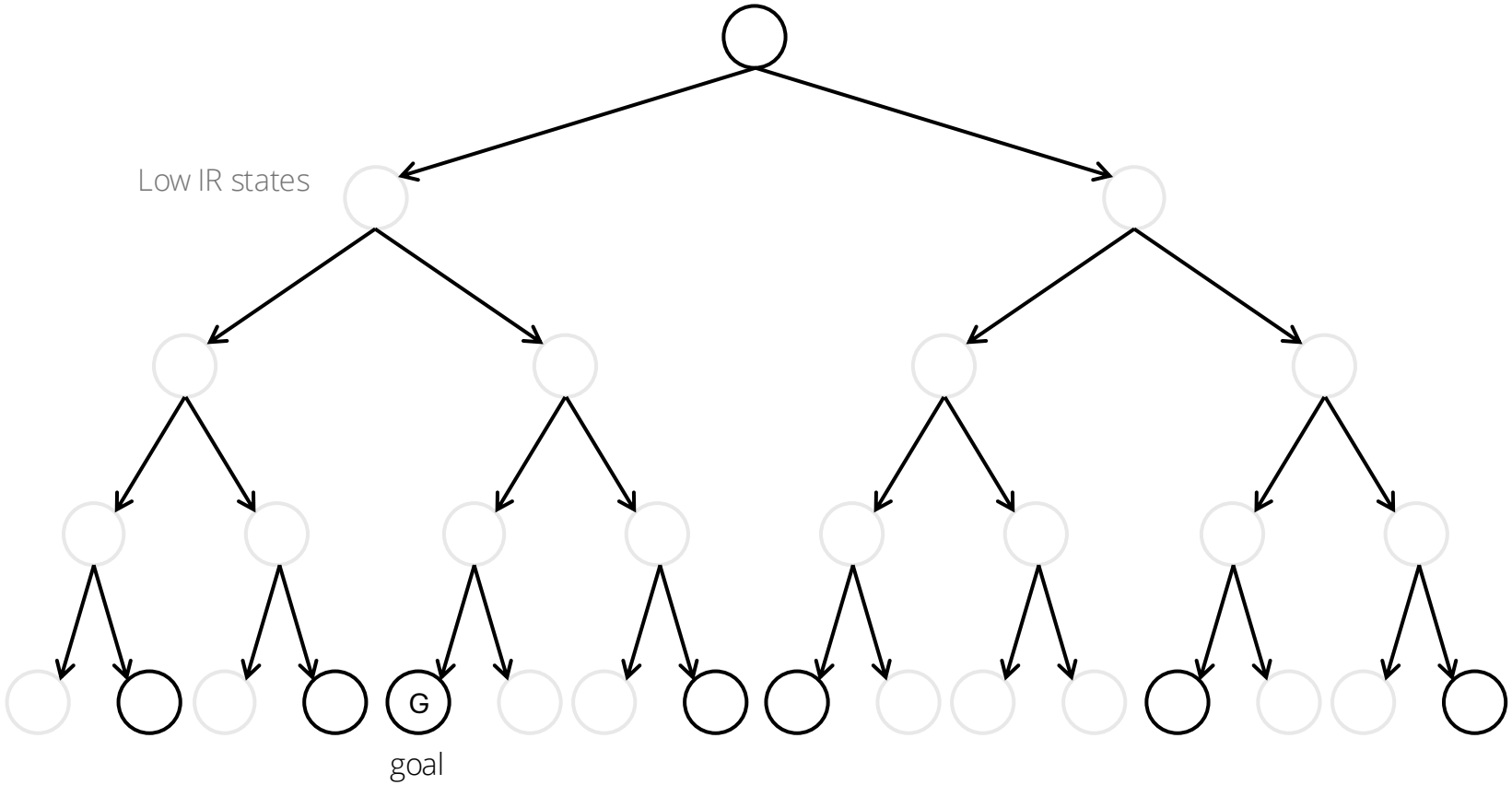
Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Previous Exploration Strategy



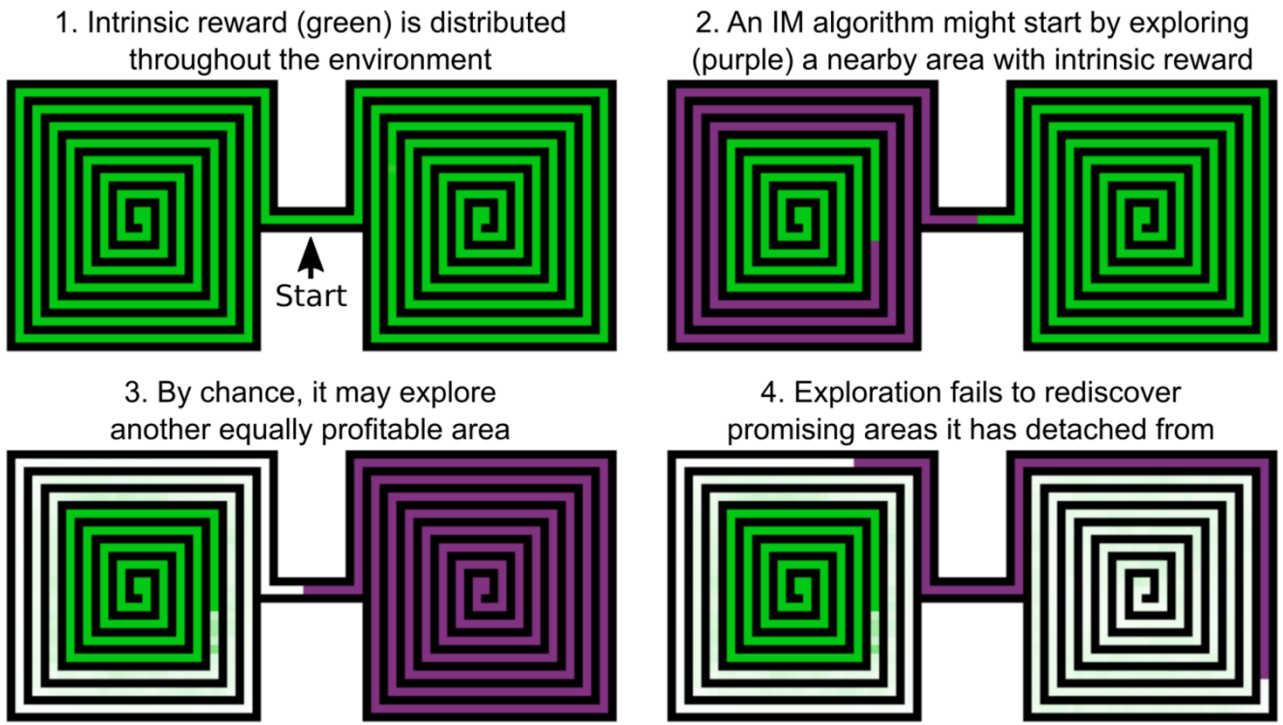
Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Previous Exploration Strategy: Little Incentive to Explore Visited States



Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Detachment



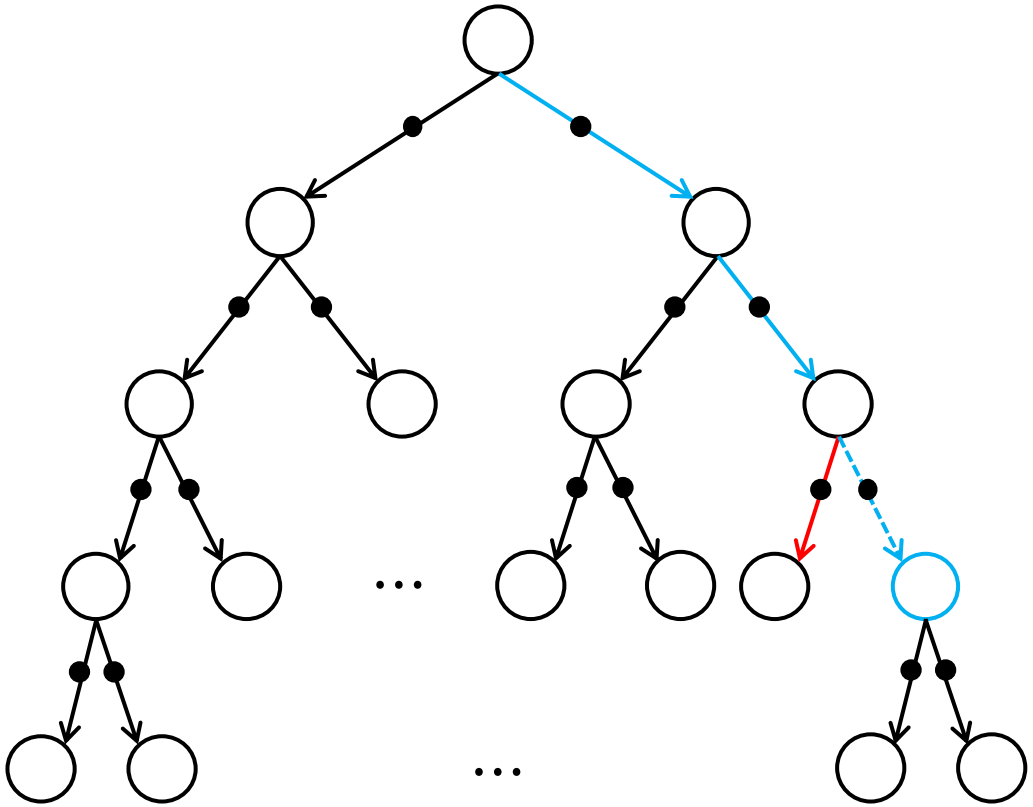
- IM-based exploration occurs as follows:

1. An agent is encouraged by discover regions with high intrinsic rewards (IR)
2. After exhausting IRs, the agent may by chance start consuming IR in another regions
3. Even if the IR in 2. is consumed, it's still hard to rediscover region of 1., where IR is also exhausted

- This phenomena is called *detachment*
- Large replay buffer may solve this issue, but why not explicitly store these previously discovered states?

Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Derailment

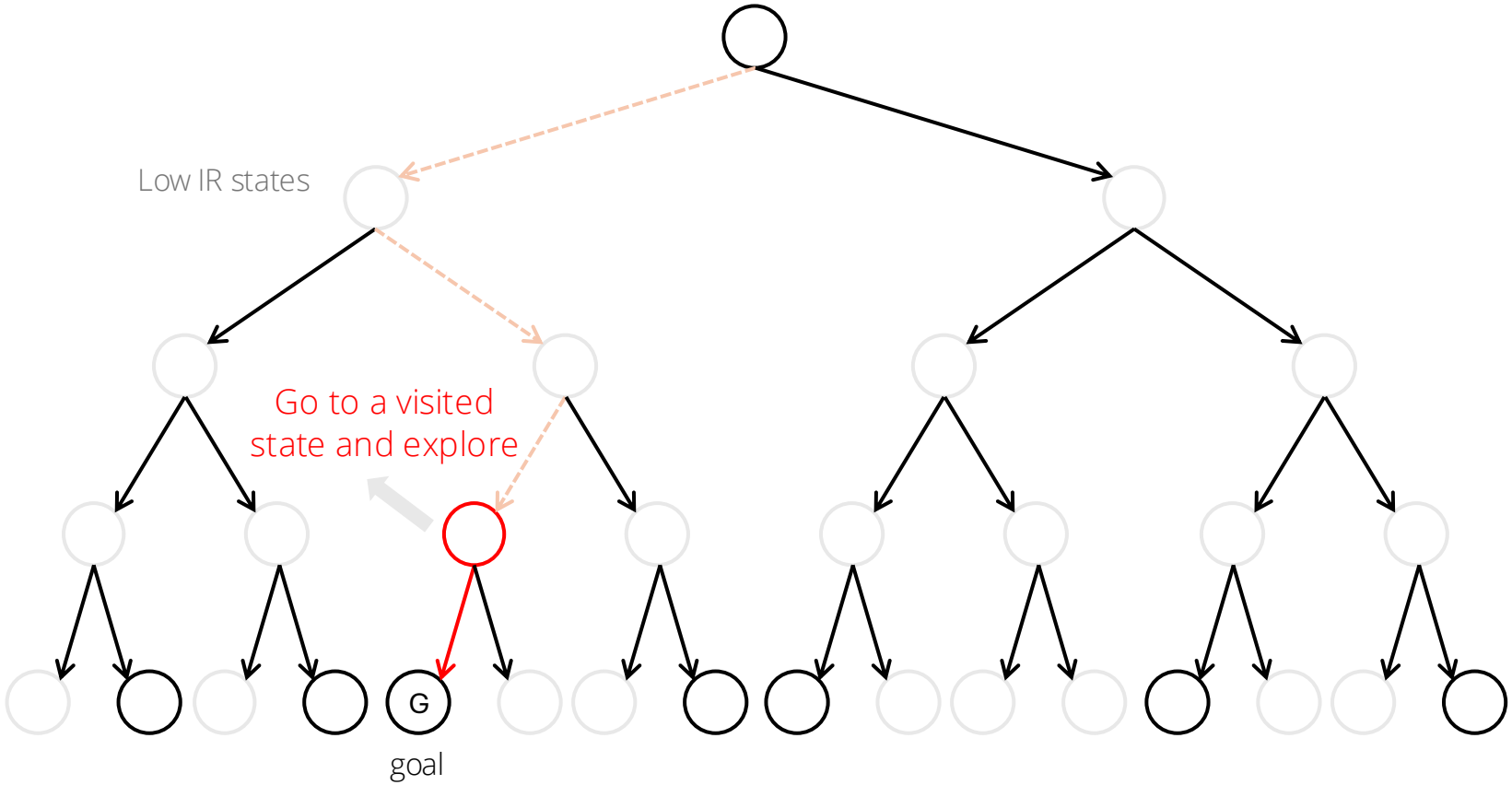


- An agent has two types of exploration mechanism:
  1. high-level exploration: IR incentive that rewards when new states are reached
  2. low-level exploration: epsilon-greedy exploration, action-space noise, or parameter-space noise

- The longer, more complex, and more precise a sequence of actions needs to be in order to reach a previously-discovered high-IR state, the more likely it is that such stochastic perturbations will "*derail*" the agent from ever returning to that state

Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Go-Explore: **Go** to a State You Have Visited without adding exploration and Randomly **Explore** from There



- Go-Explore assumes the environment can be reset to any state; in reality, you may need to train a goal-conditioned policy

Go-Explore: a New Approach for Hard-Exploration Problems. Ecoffet et al.

Go-Explore: **Go** to a State You Have Visited without adding exploration and Randomly **Explore** from There

- State representations of Go-Explore: (1) convert an image to grayscale, (2) down-scale to 11×8 resolution, and (3) rescale pixel intensities between 0 and 8, instead of the original 0 to 255

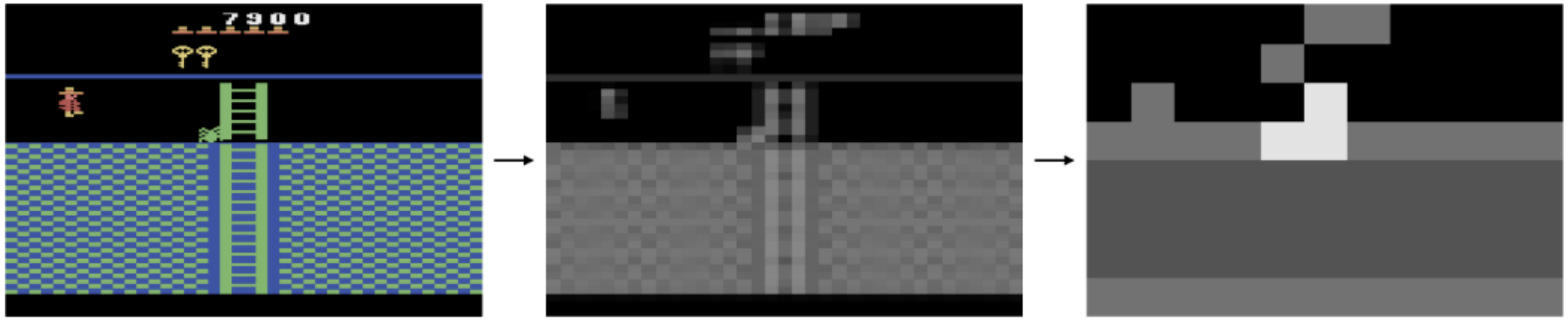


Figure 3: **Example cell representation without domain knowledge, which is simply to down-sample each game frame.** The full observable state, a color image, is converted to grayscale and downscaled to an 11×8 image with 8 possible pixel intensities.

Go-Explore: **Go** to a State You Have Visited without adding exploration and Randomly **Explore** from There

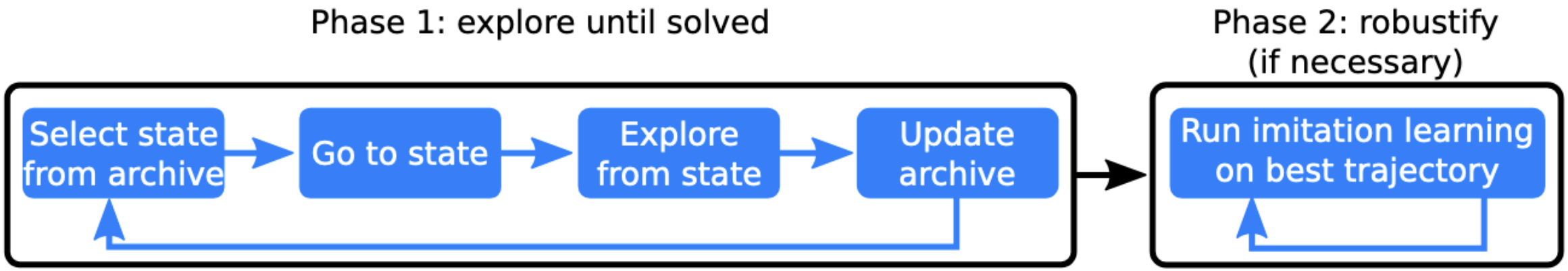
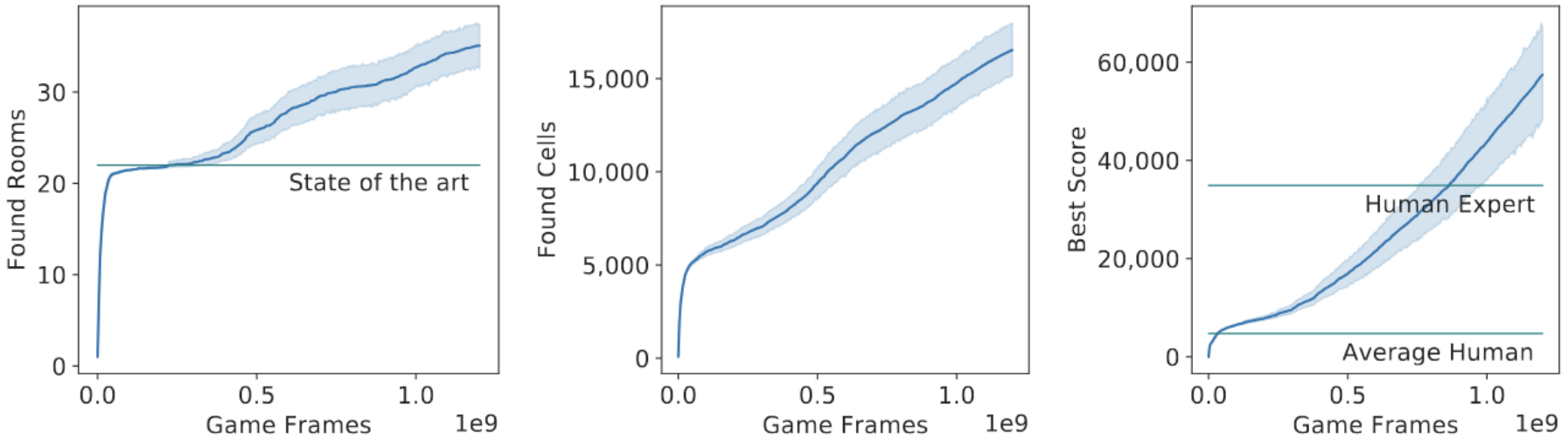


Figure 2: **A high-level overview of the Go-Explore algorithm.**

Breakthrough in Long-horizon Tasks



(a) Number of rooms found (b) Number of cells found (c) Maximum score in archive

Figure 4: **Performance of the exploration phase of Go-Explore with downscaled frames on Montezuma's Revenge.** Lines indicating human and the algorithmic state of the art are for comparison, but recall that the Go-Explore scores in this plot are on a deterministic version of the game (unlike the post-Phase 2 scores presented in this section).

Breakthrough in Long-horizon Tasks

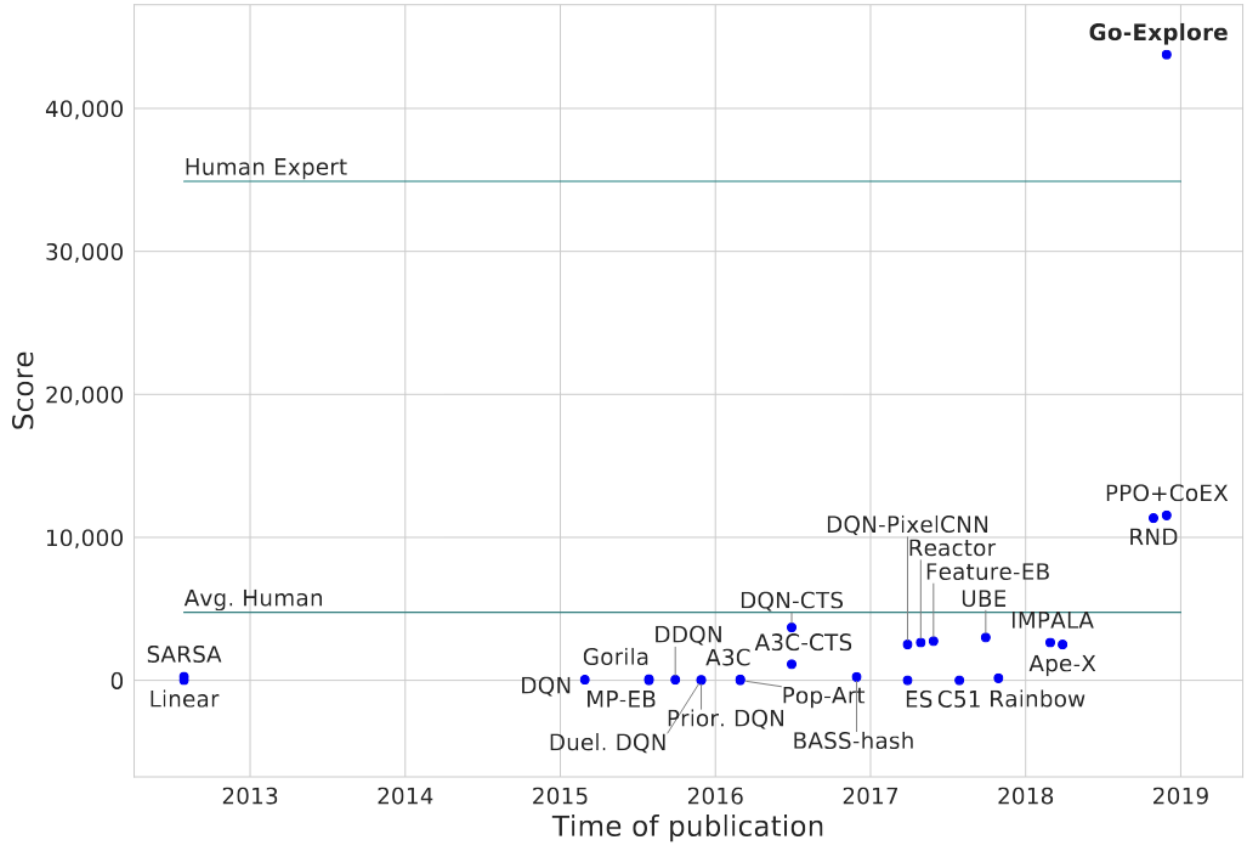


Figure 6: **History of progress on Montezuma's Revenge vs. the version of Go-Explore that does not harness domain knowledge.** Go-Explore significantly improves on the prior state of the art. These data are presented in tabular form in Appendix A.9.

How to Model Exploration?

- We can re-formulate the objective of reinforcement learning:

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extrinsic reward (related to the task)

intrinsic reward (unrelated to the task)

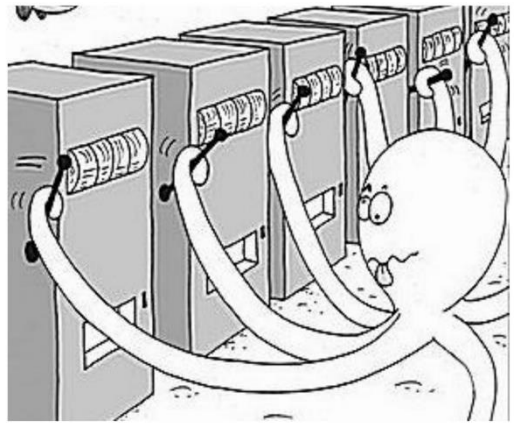
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  4. Reachability exploration
  5. Posterior sampling for exploration

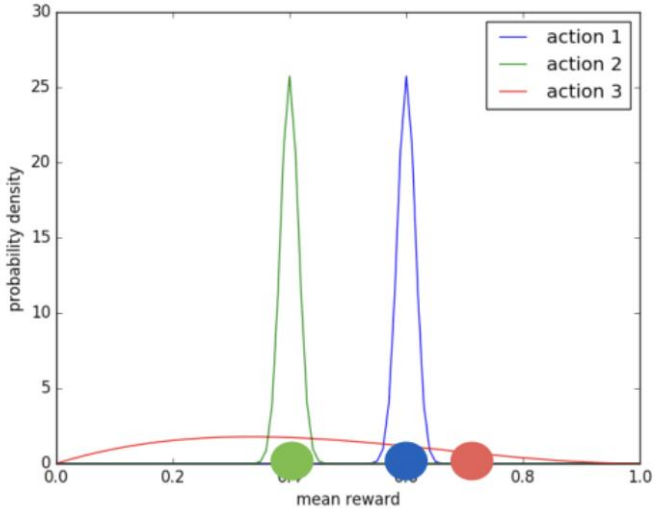
Posterior Sampling for Exploration

- Exploration often waste interaction budget on low-reward states
- Idea:
  - To explore more efficiently, an agent should explore only when there are valuable learning opportunities
  - Since any action may have long term consequences, the agent should reason about the informational value of possible observation sequence

- **Thompson sampling automatically balances exploitation and exploration** : it chooses the action that maximizes the expected reward with respect to a randomly drawn belief, and updating the belief with respected to received reward

Thompson Sampling in Multi-arm Bandits





- Posterior distribution: $P(\theta|D) \propto P(\theta)P(D|\theta)$
- Thompson sampling: Represent a posterior distribution of mean rewards of the arms, as opposed to point estimates
  1. Sample from it: $\theta_t \sim P(\theta)$
  2. Choose action: $a = \underset{a}{\text{argmax}}\, \mathbb{E}_{\theta_t}[r(a; \theta_t)]$
  3. Play action, observe reward
  4. Update the posterior distribution

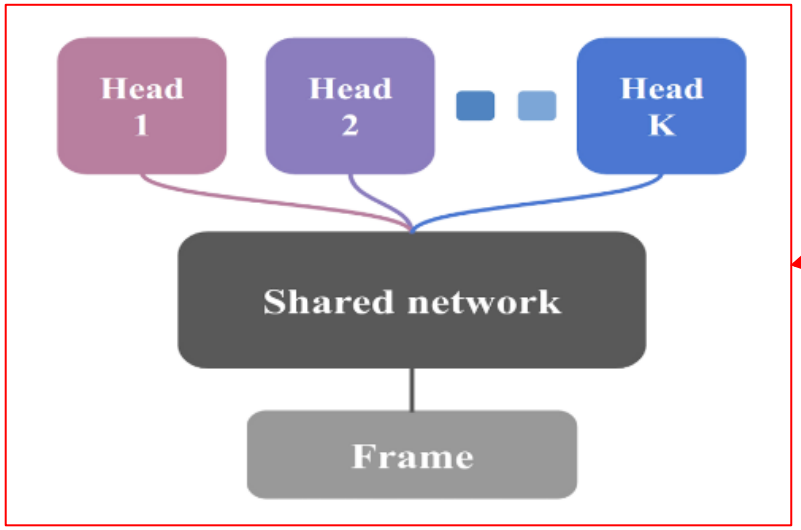
     $$P(\theta) \leftarrow P(\theta|D) \propto P(\theta)P(D|\theta)$$
- A simple explanation on stack overflow: https://stats.stackexchange.com/questions/187059/what-is-thompson-sampling-in-laymans-terms

Thompson Sampling for General MDP

- Represent a posterior distribution of Q, instead of a point estimate:
  1. Sample from $\mathbf{Q} \sim P(Q|D)$
  2. Choose action according to this Q function for an episode: $a = \underset{a}{\mathbf{argmax}}\, Q(a, s)$
  3. Play action, observe reward
  4. Update the Q distribution

- There's no need for ϵ-greedy for exploration!
- How can we learn a distribution of Q functions $P(Q|D)$ if Q function is a deep neural network?

The Same Idea of Bootstrap!

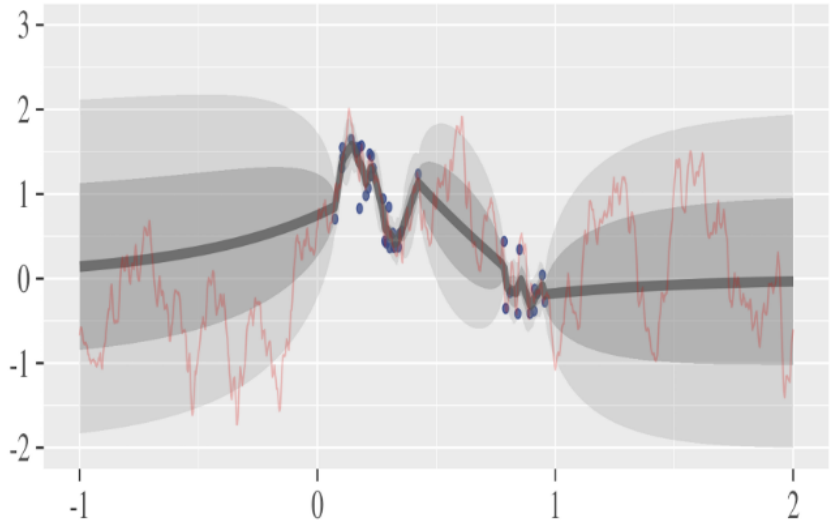


- How to represent a posterior distribution of Q?
  1. Bayesian neural networks. Estimate posteriors for the neural weights, as opposed to point estimates.
  2. Neural network ensembles. Train multiple Q-function approximations each on using different subset of the data. A reasonable approximation to 1.
  3. Neural network ensembles with shared backbone. Only the heads are trained with different subset of the data. A reasonable approximation to 2 with less computation.
  4. Ensembling by dropout. Randomly mask-out (zero out)neural network weights, to create different neural nets, both at train and test time. reasonable approximation to 2.

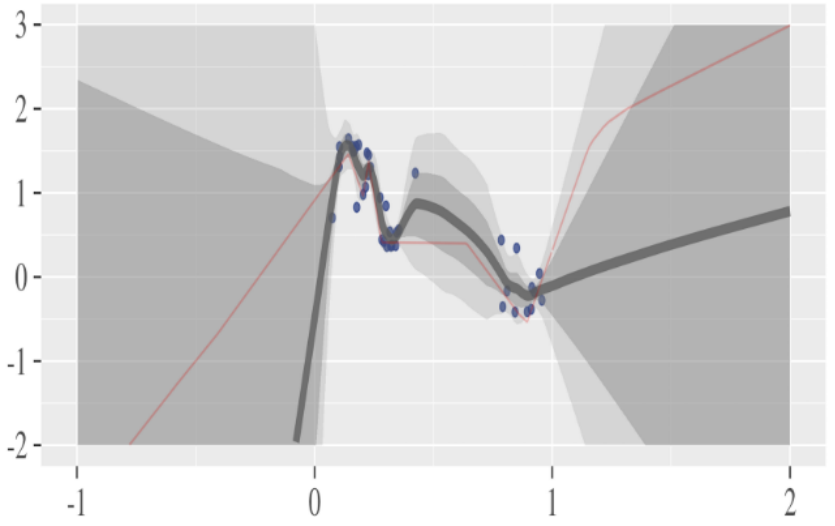
Deep Exploration via Bootstrapped DQN. Osband et al.

Slide credit K. Fragkiadaki

Bootstrap Captures the Model Uncertainty



(b) Gaussian process posterior

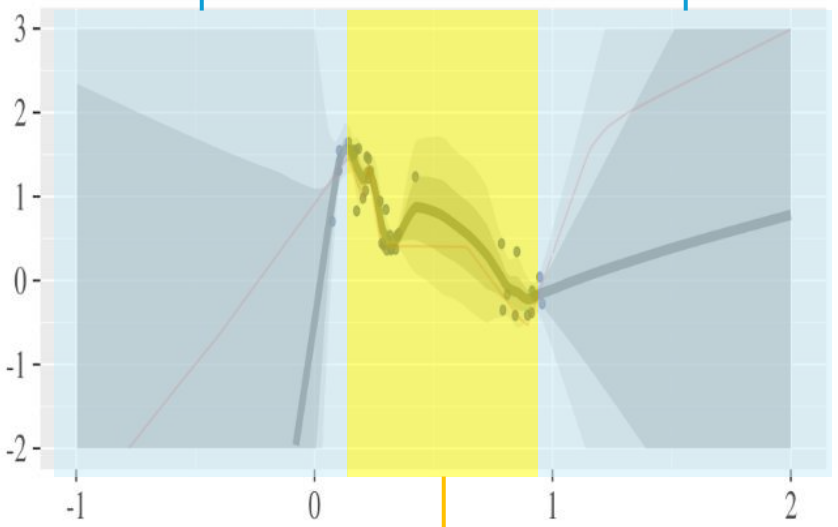


(c) Bootstrapped neural nets

Deep Exploration via Bootstrapped DQN. Osband et al.

Slide credit K. Fragkiadaki

Bootstrap Captures the Model Uncertainty



In no data regime, Q function values will have high entropy and encourage exploration

When Q values have low entropy, the agent exploit, no need for exploration

Deep Exploration via Bootstrapped DQN. Osband et al.

Slide credit K. Fragkiadaki

Results

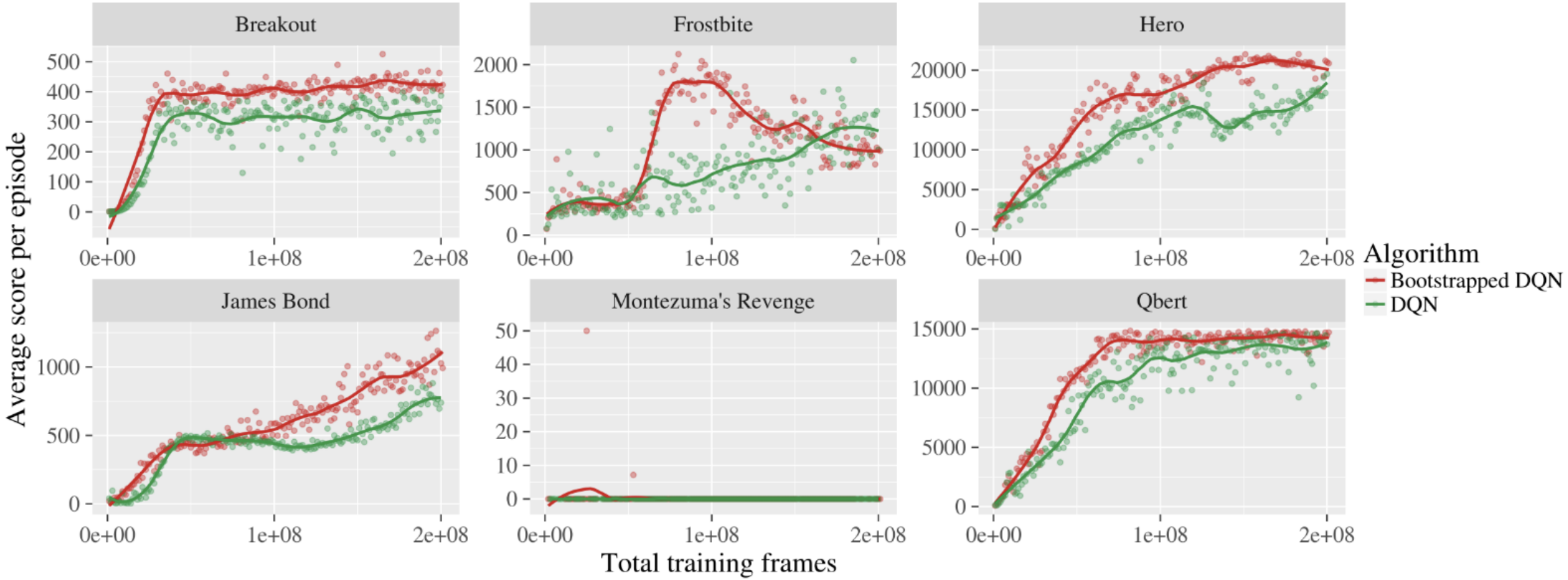
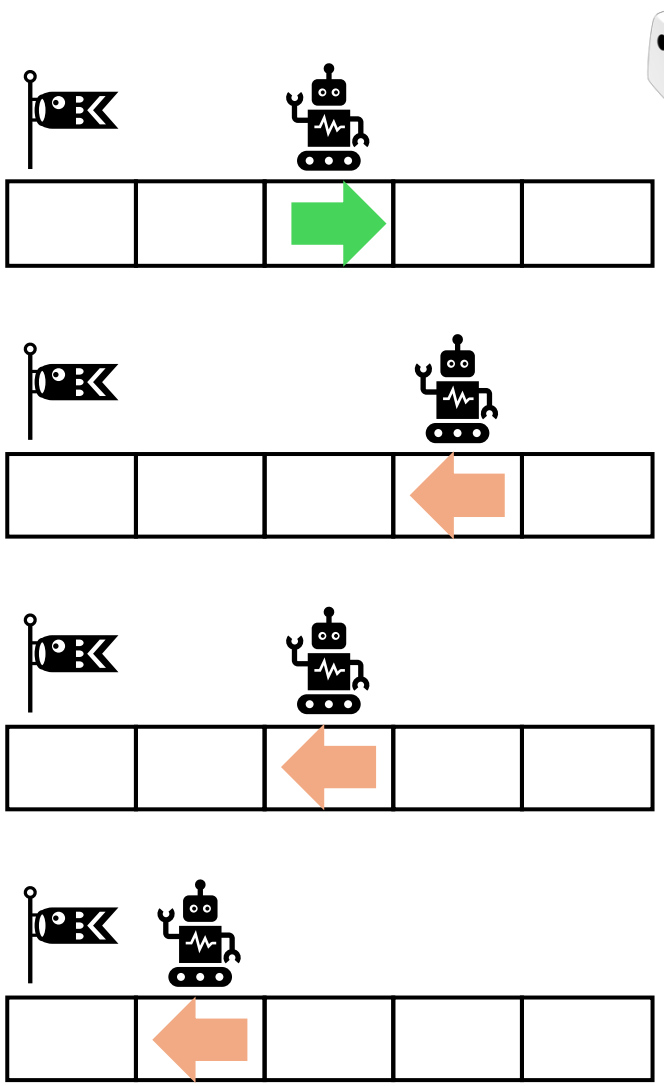


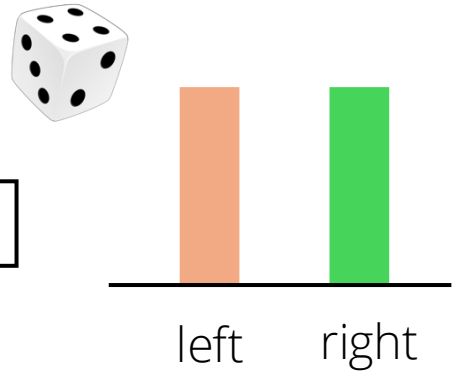
Figure 6: Bootstrapped DQN drives more efficient exploration.

Deep Exploration via Bootstrapped DQN. Osband et al.

Is Exploration Solved?

Exploration Based on Inductive Bias are Inefficient

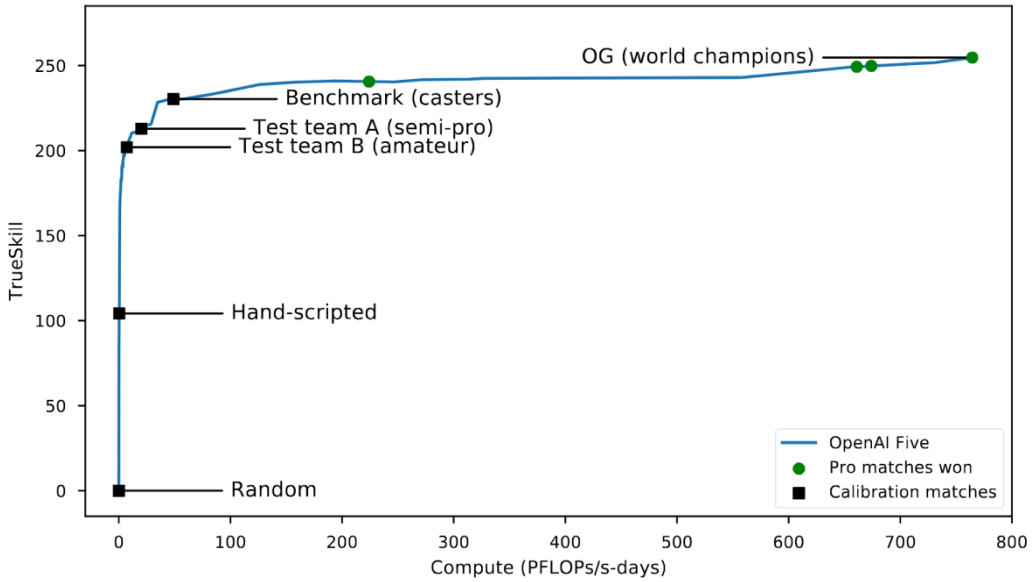




Wasting steps on obvious useless actions

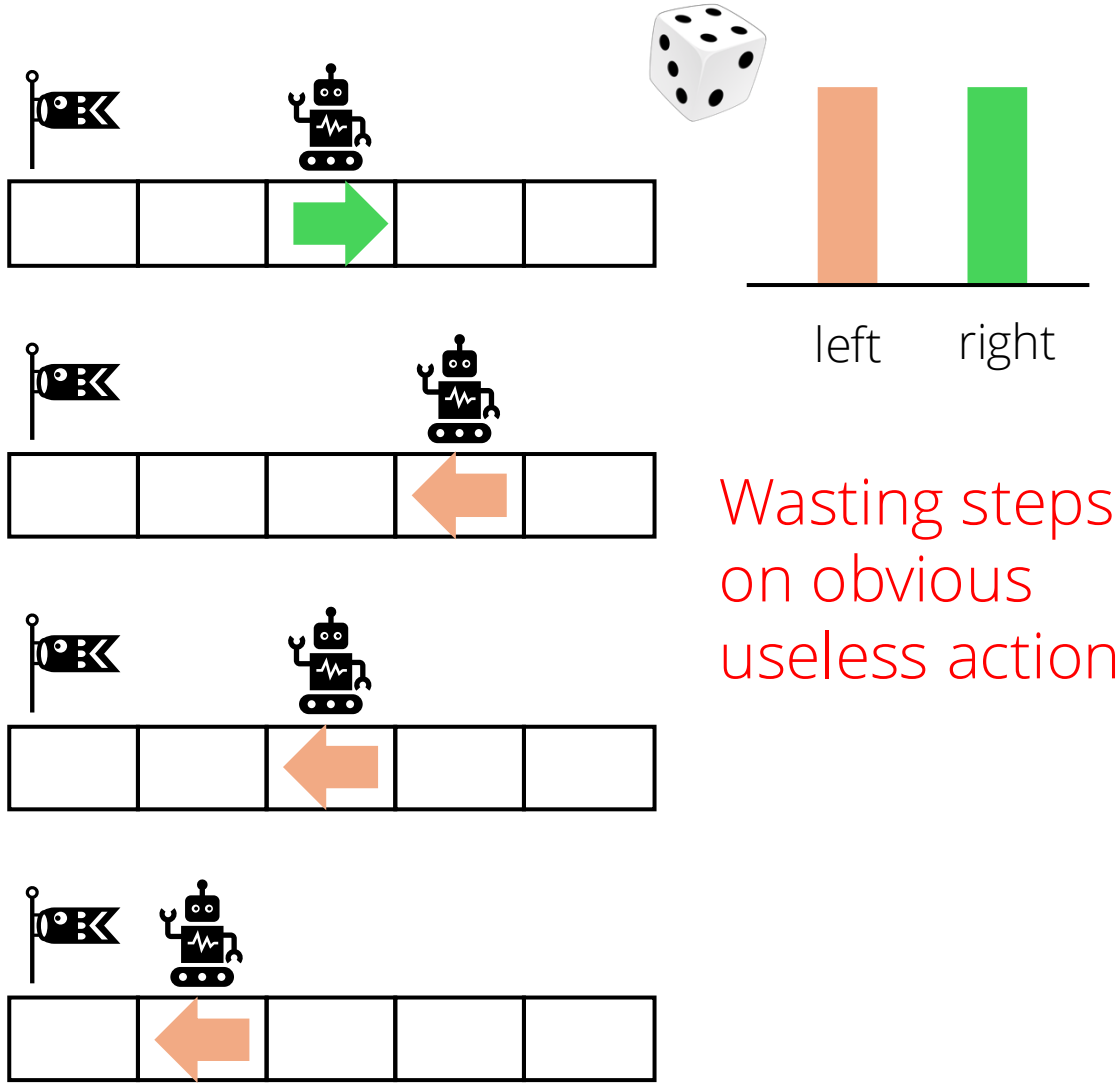




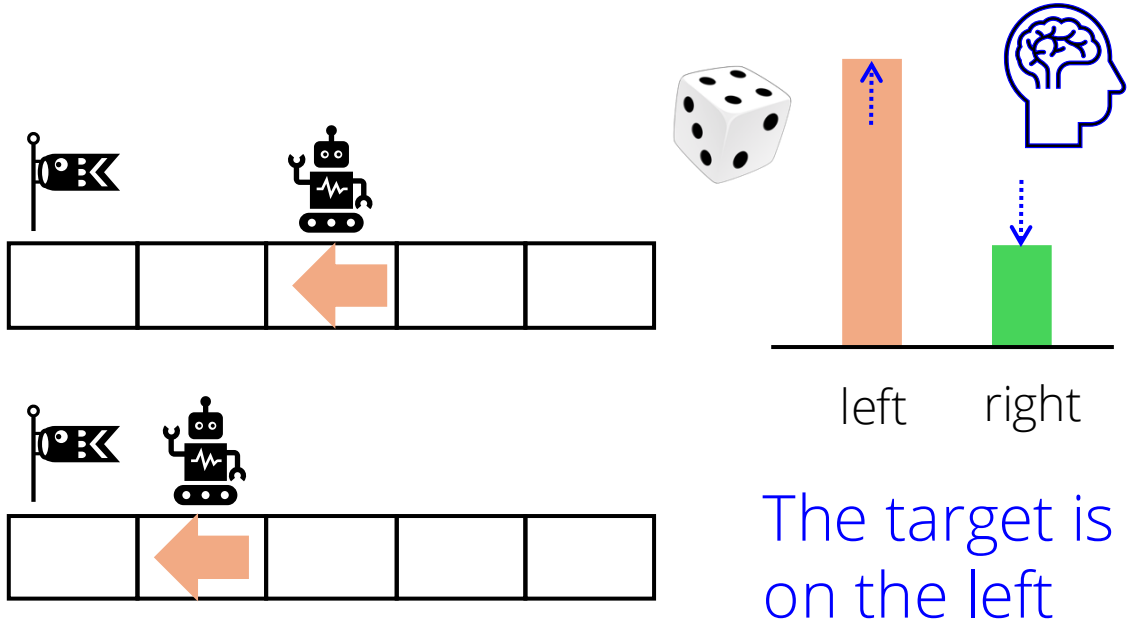


10 months

Exploration Based on Inductive Bias vs. Semantic Priors



Wasting steps on obvious useless actions



The target is on the left

Where to Obtain Semantic Priors? Human Knowledge!

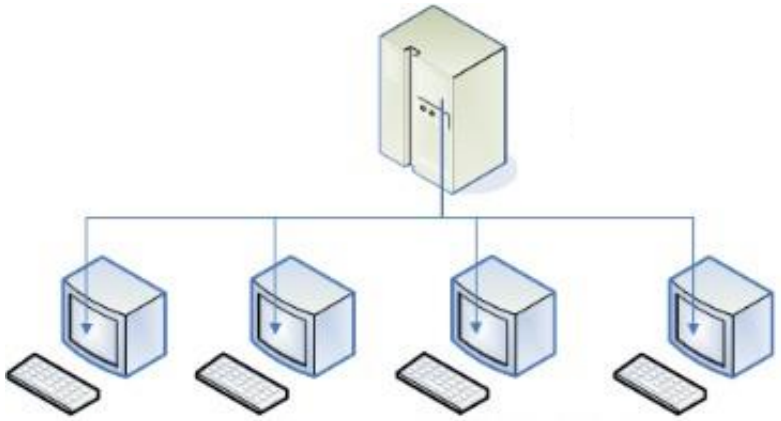


Image credit 鳥哥的首頁

Agentic Robotics

Visual imitation learning

